
Multivariate Time Series Forecasting needs Cross Variable Loss

Anonymous Author(s)

Affiliation

Address

email

Abstract

Multivariate time series forecasting presents unique challenges because future variables often co-evolve under shared system dynamics. While existing studies mainly focus on cross-variable dependencies in historical observations, dependencies among future values are much less explored. Specifically, modern forecasting models largely follow the Direct Forecasting (DF) paradigm, generating multi-step forecasts with point-wise objectives that do not explicitly constrain cross-variable structure. In this work, we show that the DF objective is mismatched in the presence of cross-variable and lagged dependencies, revealing an objective gap. To address this issue, we propose **Cross-Variable Loss (CvLoss)**, a plug-in structural regularizer that constrains forecast residuals on a cross-variable graph. CvLoss penalizes inconsistent edge-wise residual differences over forecast patches, encouraging consistency across both synchronous and asynchronous interactions. Our experiments show that CvLoss consistently improves competitive forecasting models, outperforms representative learning objectives, and is compatible with a variety of forecasting backbones. Code is available at <https://anonymous.4open.science/r/CvLoss>.

1 Introduction

Multivariate time series forecasting (MTSF) is a fundamental task across diverse domains, where the core challenge lies in accurately capturing complex synergistic relationships among variables. To model these temporal dynamics, recent pioneering works [Liu et al., 2024, Zhao and Shen, 2024, Qiu et al., 2025a, Hu et al., 2025a] have predominantly focused their efforts on extracting representations from historical input features, a paradigm known as input-side channel dependence modeling. In stark contrast, the intrinsic correlations among future predictions at the output-side have been largely overlooked. Current mainstream Direct Forecasting (DF) paradigms Wang et al. [2025a] typically decouple the multivariate outputs forcibly, projecting the unified historical embeddings independently into future time steps, thereby degrading MTSF into a collection of isolated scalar prediction problems. This asymmetry heavily prioritizes the input while neglecting the output. Consequently, it not only severs the physical co-evolution among future variables but also induces severe structural errors.

On one hand, blindly decoupling prediction targets directly violates the intrinsic physical and operational laws of multivariate systems. In the real world, time series are rarely generated independently; rather, they are governed by shared system dynamics and exhibit strong co-evolutionary characteristics. For instance, in quantitative financial markets [Hu et al., 2025b], the returns of correlated assets often display highly synchronous surges, plunges, or hedging effects when subjected to macroeconomic shocks Hu et al. [2025c]. Similarly, in traffic flow forecasting, the traffic volumes recorded by hundreds of sensors within an urban network are driven by shared meteorological conditions (e.g., sudden rainstorms causing widespread congestion), and synchronized daily commuting patterns. This results in a profound co-evolution of future traffic dynamics, as illustrated in Figure 5. If a model

isolates these variables at the output stage, the predicted curves for individual variables might appear numerically well-fitted. However, when combined into a joint multivariate forecast, they frequently produce contradictory results that violate true physical constraints and market rules.

Furthermore, overlooking these synergistic relationships induces critical structural errors at the optimization level. Modern MTSF models rely heavily on point-wise loss functions, such as Mean Squared Error (MSE) Wang et al. [2024], Qiu et al. [2026], for optimization. From the perspective of probabilistic inference, this point-wise optimization is fundamentally predicated on a highly restrictive assumption: the prediction residuals across all variables and time steps follow completely independent and isotropic Gaussian distributions (i.e., possessing a diagonal covariance matrix). However, given the aforementioned co-evolutionary phenomena, the true spatio-temporal covariance matrix of the residuals inevitably contains numerous non-zero off-diagonal elements. This creates an unbridgeable theoretical divide, termed an Objective Gap, between existing standard loss functions and the true joint spatio-temporal distribution of the data. Consequently, even if two models achieve remarkably similar point-wise MSE, their cross-variable covariance structures might differ drastically, completely failing to capture the joint geometric topology of the original data.

To overcome these dual physical and mathematical deficiencies, the optimization objective must transcend the singular pursuit of absolute point-wise accuracy and explicitly constrain the relative structural consistency among variables. This implies that we must capture not only the concurrent interactions (synchronous effects) between variables at the same time step but also the lagged propagation of system perturbations as time evolves (asynchronous effects). To elegantly unify these two phenomena, we introduce a graph-structured perspective, as graphs are inherently suited for depicting complex, non-uniform pair-wise interactions among multivariate data. Specifically, we propose a novel graph-based structural regularization scheme termed Cross-Variable Loss (CvLoss). By constructing a cross-variable graph, CvLoss connects different variables at identical time steps to capture synchronous concurrent interactions, while also linking variables across different steps to model asynchronous lagged effects. Essentially, CvLoss imposes an ℓ_1 -norm Graph Total Variation (Graph TV) regularization on the prediction residual field, compelling the model to strictly adhere to the underlying relative structural consistency while minimizing absolute errors. Furthermore, as a plug-and-play regularizer, CvLoss seamlessly integrates into various mainstream forecasting architectures. Our introduced edge-sampling mechanism significantly reduces computational complexity during training, and crucially, the regularizer imposes zero overhead during inference, ensuring testing speed remains completely unaffected while maintaining extremely high predictive accuracy.

Our main contributions are summarized as follows:

- We reveal a critical limitation in current MTSF paradigms, demonstrating that standard point-wise learning objectives neglect inherent inter-channel error dependencies among future predictions.
- We propose a novel Cross-Variable Loss acting as a structural regularizer, explicitly constraining the inter-channel relationships of predicted values via a residual consistency formulation.
- Extensive experiments demonstrate our proposed objective consistently improves various state-of-the-art models, achieving superior predictive performance with affordable computational overhead.

2 Preliminaries and Related Work

2.1 Problem definition

In this paper, we focus on the multivariate time series forecasting (MTSF) problem. We use uppercase letters (e.g., X) to denote matrices and lowercase letters (e.g., x) to denote scalars. Given a multivariate time-series dataset with D variables, the input historical sequence is defined as $X \in \mathbb{R}^{H \times D}$, and the ground-truth label sequence is defined as $Y \in \mathbb{R}^{T \times D}$, where H is the lookback window and T is the forecast horizon. Modern time series forecasting models are primarily trained under a multitask learning manner, known as the direct forecasting (DF) paradigm. Therefore, the target is to learn a multi-output predictor $f_\theta : \mathbb{R}^{H \times D} \rightarrow \mathbb{R}^{T \times D}$ that directly produces T -step forecasts for all D variables simultaneously, yielding the prediction $\hat{Y} = f_\theta(X)$. Unlike current paradigms that decouple multivariate outputs into isolated scalar predictions, our formulation explicitly maintains the joint structure to capture cross-channel dependencies and shared temporal dynamics.

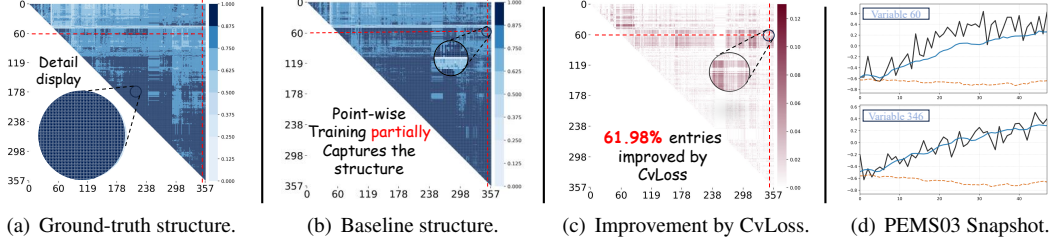


Figure 1: Motivating example on PEMS03 with input length $H = 96$ and forecast horizon $T = 48$. (a,b) Cross-variable correlation structures of the ground truth and the plain iTransformer [Liu et al., 2024] prediction. (c) Entries where CvLoss reduces structural error over the baseline, covering 61.98% of cross-variable pairs. (d) Forecast snapshot: orange denotes the baseline prediction, black denotes the ground truth, and blue denotes ours (CvLoss). See details in Appendix B.3.

2.2 Learning objectives in time-series forecasting

Recent learning objectives extend point-wise MSE to capture temporal dynamics through shape alignment [Liu et al., 2021, Le Guen and Thome, 2019, Cuturi and Blondel, 2017], likelihood maximization [Wang et al., 2026a, 2025b, Lange et al., 2021, Wang et al., 2025a], distribution balancing [Pan et al., 2026, Wang et al., 2026b], and intra-sequence modeling via decomposition [Qiu et al., 2025b] or local step alignment [Xiong et al., 2025, Kudrat et al., 2025]. Despite these advancements, existing approaches fundamentally treat multivariate forecasting as a collection of isolated scalar predictions. By optimizing exclusively for intra-channel temporal structures, they completely neglect explicit constraints on cross-variable dependencies. Consequently, these methods fail to capture the joint geometric structure and correlated disturbances among multivariate outputs.

2.3 Cross-variable Dependencies in time-series forecasting

Effectively modeling cross-variable dependencies is fundamental for multivariate time-series forecasting (MTSF) [Wang et al., 2024, Liang et al., 2024]. Existing paradigms typically employ Channel Dependence (CD) strategies to explicitly capture inter-channel correlations, as seen in representative architectures like iTransformer [Liu et al., 2024], TSMixer [Chen et al., 2023], Crossformer [Zhang and Yan, 2023], and TimesNet [Wu et al., 2023]. However, CD models often overfit to spurious correlations and struggle with noise interference from irrelevant variables. To address this, recent advancements introduce Channel Partiality (CP) to selectively filter dependencies. For instance, DUET [Qiu et al., 2025a] proposes a Channel Clustering Module to filter noisy channels via frequency-domain soft clustering. Similarly, TimeFilter [Hu et al., 2025a] constructs a patch-specific spatial-temporal graph with dynamic routing to retain essential correlations while discarding irrelevant noise. Other approaches like MTGNN [Wu et al., 2020] and MCformer [Han et al., 2024a] also explore sparse interactions to balance accuracy and efficiency. Despite these significant improvements, current models predominantly focus on extracting representations from historical observations [Zhao et al., 2024, Wei et al., 2026]. The complex structural geometry and co-evolving cross-variable dependencies among future predictions remain fundamentally underexplored.

3 Proposed Method

3.1 Motivation

Modern time series forecasting models are primarily trained under the multitask learning manner, known as the direct forecasting (DF) paradigm [Wang et al., 2025a]. Given an input history $X \in \mathbb{R}^{H \times D}$, a multi-output predictor $f_\theta : \mathbb{R}^{H \times D} \rightarrow \mathbb{R}^{T \times D}$ directly produces T -step forecasts for D variables: $\hat{Y} = f_\theta(X)$. The standard training objective is the point-wise Mean Squared Error (MSE):

$$\mathcal{L}_{\text{df}} = \frac{1}{T} \sum_{t=1}^T \frac{1}{D} \|Y_{t,:} - \hat{Y}_{t,:}\|_2^2 \quad (1)$$

121 This objective is simple and effective, but it only penalizes prediction errors at individual time-variable
 122 entries. Although modern forecasting architectures may implicitly capture cross-variable interactions
 123 through shared parameters, such dependencies are not explicitly reflected in the objective in Eq. (1),
 124 which does not constrain the *relative error structure* across variables. As a result, two models with
 125 similar point-wise MSE may still exhibit substantially different cross-variable correlated patterns.

126 From a broader perspective, multivariate forecasting is not merely a collection of isolated scalar
 127 prediction problems, since different variables often evolve with shared dynamics and correlated
 128 disturbances. This view is also consistent with multi-task learning, where explicit coupling across
 129 related targets can improve generalization [Evgeniou et al., 2005]. Consequently, relying purely on
 130 point-wise estimation may fail to fully exploit the shared structure across variables.

131 Motivated by these perspectives, we seek to augment the vanilla DF objective with an explicit
 132 structural regularizer that captures dependencies among variables. Rather than replacing point-wise
 133 supervision, this additional term is intended to complement it by encouraging consistency in the
 134 residual structure across related variables. To clarify why such a structural term may be beneficial,
 135 we first examine the standard DF objective from a probabilistic perspective.

136 Let $\mathbf{e} = \text{vec}(\hat{Y} - Y) \in \mathbb{R}^{TD}$ denote the flattened prediction error vector over the entire forecast
 137 horizon and all variables. Suppose the residuals follow a zero-mean multivariate Gaussian distribution
 138 $\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \Sigma_{ST})$, where $\Sigma_{ST} \in \mathbb{R}^{TD \times TD}$ is the spatiotemporal covariance matrix that captures both
 139 cross-variable and cross-temporal dependencies.

140 **Theorem 3.1 (Objective Gap with Spatiotemporal Dependencies)** *Ignoring additive constants*
 141 *independent of the predictions, the scaled negative log-likelihood (NLL) under the true covari-*
 142 *ance Σ_{ST} is $\mathcal{L}_{\text{nll}} = \frac{1}{2TD} \mathbf{e}^\top \Sigma_{ST}^{-1} \mathbf{e}$. On the other hand, the standard DF objective in Eq. (1) can be*
 143 *written as $\mathcal{L}_{\text{df}} = \frac{1}{TD} \mathbf{e}^\top \mathbf{e}$. Therefore, compared with the isotropic Gaussian NLL under variance σ^2 ,*
 144 *the discrepancy between the two objectives is*

$$\Delta = \mathcal{L}_{\text{nll}} - \frac{1}{2\sigma^2} \mathcal{L}_{\text{df}} = \frac{1}{2TD} \mathbf{e}^\top \left(\Sigma_{ST}^{-1} - \frac{1}{\sigma^2} \mathbf{I} \right) \mathbf{e}, \quad (2)$$

145 where $\mathbf{I} \in \mathbb{R}^{TD \times TD}$ is the identity matrix. In real-world multivariate time series, the presence
 146 of cross-variable and lagged dependencies ensures that the true precision matrix Σ_{ST}^{-1} contains
 147 non-zero off-diagonal elements. Consequently, $\Sigma_{ST}^{-1} \neq \frac{1}{\sigma^2} \mathbf{I}$, meaning a strictly non-zero objective
 148 gap ($\Delta \neq 0$) inherently exists because the standard point-wise DF objective neglects these underlying
 149 structural correlations.

150 Theorem 3.1 shows that the standard point-wise DF objective coincides with the isotropic Gaussian
 151 NLL only when the residual covariance is spherical, i.e., $\Sigma_{ST} = \sigma^2 \mathbf{I}$. When substantial cross-
 152 variable dependencies are present, potentially together with lagged temporal interactions, minimizing
 153 point-wise MSE alone does not explicitly account for the underlying residual dependency structure.
 154 However, in real-world systems, dynamic physical processes often induce lagged effects—such as a
 155 disturbance in one variable propagating to another variable after several time steps. Consequently,
 156 off-diagonal elements in Σ_{ST} , which represent cross-variable and cross-temporal correlations, are
 157 significantly non-zero. As a result, point-wise MSE may fail to reflect the true spatiotemporal error
 158 geometry and may not fully exploit correlated residual structure during training.

159 3.2 Predicting Multivariate Time Series with Cross Variable Loss Constraints

160 3.2.1 Definition

161 **Patch nodes.** Let $Y \in \mathbb{R}^{T \times D}$ be the target and $\hat{Y} = f_\theta(X) \in \mathbb{R}^{T \times D}$ be the prediction. We split
 162 the time axis into P non-overlapping patches of length L , so that $T = P \times L$. Each node is defined as
 163 a pair $v = (p, d)$, where $p \in \{1, \dots, P\}$ indexes the patch and $d \in \{1, \dots, D\}$ indexes the variable.
 164 Let $z_v \in \mathbb{R}^L$ and $\hat{z}_v \in \mathbb{R}^L$ denote the ground-truth and predicted patch vectors at node v , respectively.
 165 Stacking all nodes gives

$$z \in \mathbb{R}^{N \times L}, \quad \hat{z} \in \mathbb{R}^{N \times L}, \quad N = P \times D. \quad (3)$$

166 We define the node-wise residuals as

$$e_v = \hat{z}_v - z_v \in \mathbb{R}^L, \quad (4)$$

167 and stack them into the residual matrix $e \in \mathbb{R}^{N \times L}$. For $L = 1$, patch nodes recover point nodes.

Cross-variable edges. To capture the spatiotemporal dependencies across different series, we introduce an edge set $\mathcal{E}$ to define the cross-variable graph topology. Guided by Theorem 3.1, rather than explicitly estimating the intractable precision matrix, we utilize $\mathcal{E}$ as a practical surrogate structure to regularize residual discrepancies. This formulation is highly flexible and can easily accommodate different structural priors. Specifically, we design three variants for constructing $\mathcal{E}$: (1) *Synchronous cross-variable edges*, which strictly capture concurrent interactions at the same time step (i.e., connecting identical patch indices); (2) *Asynchronous cross-variable edges*, which exclusively model lagged dependencies across different time steps; and (3) *Fully connected cross-variable edges*, which include both synchronous and lagged interactions. We adopt the fully connected graph as our default setting to maximize the coverage of potential interactions without relying on domain-specific prior knowledge. We evaluate these topological designs in the ablation studies in Section 4.5.

Difference operator. Fix an arbitrary orientation for each undirected edge and let $A \in \mathbb{R}^{|\mathcal{E}| \times N}$ be the incidence matrix: for an edge $k = (i, j)$, the k -th row contains $+1$ at position i , -1 at position j , and 0 elsewhere. Since each node is an L -dimensional patch, we lift the incidence operator as

$$\tilde{A} := A \otimes I_L. \quad (5)$$

Let $\text{vec}(e) \in \mathbb{R}^{NL}$ denote the vectorization of e . Then $\tilde{A} \text{vec}(e) \in \mathbb{R}^{|\mathcal{E}|L}$ stacks the residual differences $e_i - e_j \in \mathbb{R}^L$ over all edges.

3.2.2 Cross-Variable Loss

Building upon the defined graph topology, we now formalize our objective. First, by reshaping the predictions into patch matrices, the standard DF loss can be equivalently rewritten as

$$\mathcal{L}_{\text{df}} = \frac{1}{N \times L} \|\hat{z} - z\|_F^2 = \frac{1}{T \times D} \|\hat{Y} - Y\|_F^2. \quad (6)$$

To capture the relative structural geometry between variables, for each edge $(i, j) \in \mathcal{E}$, we define the predicted and true patch differences as

$$\Delta_{ij}^{\text{pred}} = \hat{z}_i - \hat{z}_j, \quad \Delta_{ij}^{\text{true}} = z_i - z_j. \quad (7)$$

We then define the Cross-Variable Loss (CvLoss) as the mean absolute discrepancy of these structural differences:

$$\mathcal{L}_{\text{cv}} := \frac{1}{|\mathcal{E}| \times L} \sum_{(i,j) \in \mathcal{E}} \|\Delta_{ij}^{\text{pred}} - \Delta_{ij}^{\text{true}}\|_1. \quad (8)$$

Using the node-wise residual $e_i = \hat{z}_i - z_i$, the structural discrepancy on each edge simplifies neatly:

$$\Delta_{ij}^{\text{pred}} - \Delta_{ij}^{\text{true}} = (\hat{z}_i - \hat{z}_j) - (z_i - z_j) = e_i - e_j. \quad (9)$$

Therefore, CvLoss admits a compact graph-theoretic form via the incidence matrix $\tilde{A}$:

$$\mathcal{L}_{\text{cv}} = \frac{1}{|\mathcal{E}| \times L} \sum_{(i,j) \in \mathcal{E}} \|e_i - e_j\|_1 = \frac{1}{|\mathcal{E}| \times L} \|\tilde{A} \text{vec}(e)\|_1. \quad (10)$$

This computes the total variation of prediction errors across the graph.

3.2.3 Interpretation

The role of CvLoss is to explicitly regularize the cross-variable residual structure that is not directly controlled by the point-wise DF loss. In particular, $\mathcal{L}_{\text{df}}$ encourages each residual e_v to be small in absolute magnitude, while $\mathcal{L}_{\text{cv}}$ encourages connected nodes to have compatible residual patterns. This induces an explicit coupling across variables at the objective level.

From the graph perspective, $\mathcal{L}_{\text{cv}}$ is an ℓ_1 -type graph total variation regularizer on the residual field. Unlike point-wise MSE, which evaluates each entry independently, CvLoss constrains the edge-wise discrepancies of prediction errors across variables. This encourages the model to exploit shared temporal structure and improves coordination among related variables.

We use the ℓ_1 norm because it is less sensitive to a few large edge-wise mismatches caused by localized anomalies or variable-specific shocks, while still promoting graph-level residual consistency.

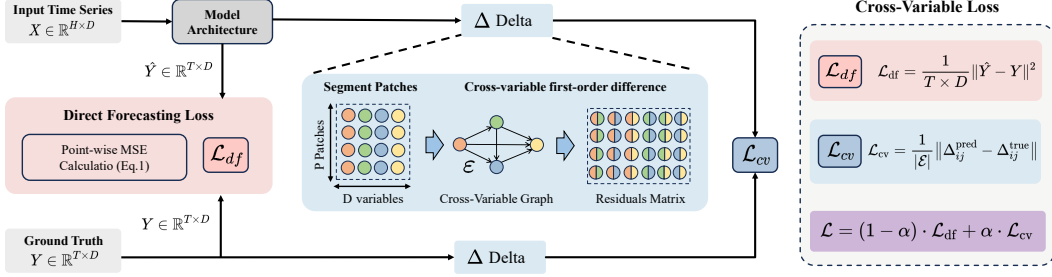


Figure 2: Overview of the workflow CvLoss.

3.3 Model Implementation

Our framework augments vanilla DF with a Cross-Variable Loss (CvLoss) constraint, complementing point-wise supervision with cross-variable regularization.

As shown in Figure 2, the normalized history X is fed into the model to produce $\hat{Y} = f_{\theta}(X)$. We compute the point-wise error $\mathcal{L}_{df}$, map forecasts and labels into patch nodes, and compute the structural error $\mathcal{L}_{cv}$ over $\mathcal{E}$. Since the incidence operations are linear, $\mathcal{L}_{cv}$ is optimized end-to-end.

Finally, the independent DF error and the structural cross-variable error are fused. A weighting parameter $\alpha \in (0, 1)$ controls the relative contribution of each objective:

$$\mathcal{L}_{\alpha} := (1 - \alpha) \cdot \mathcal{L}_{df} + \alpha \cdot \mathcal{L}_{cv}. \quad (11)$$

By sharing statistical strength across the multivariate series, our framework complements the DF paradigm with explicit cross-variable regularization while preserving its practical advantages, including efficient inference and multi-task learning capabilities. It is worth noting that the standard $\mathcal{L}_{df}$ term serves as a mathematically necessary absolute anchor. It prevents uniform drift and trivial translation-invariant solutions (where $e_i = e_j = c \neq 0$) even when the structural penalty dominates (i.e., $\alpha \rightarrow 1$). Furthermore, our proposed regularization scheme is entirely model-agnostic, seamlessly compatible with various forecasting architectures f_{θ} (e.g., Transformers, CNNs, and MLPs).

4 Experiments

To demonstrate the efficacy of CvLoss, the following aspects deserve empirical investigation:

- **Performance:** *Does CvLoss perform well?* In Section 4.2, we benchmark CvLoss against state-of-the-art baselines, and in Section 4.3, we compare it with alternative learning objectives.
- **Gain:** *Why does it work?* In Section 4.5, we perform an ablative study, dissecting the individual components of CvLoss and clarifying their contributions to forecast accuracy.
- **Generality:** *Does it support other models?* In Section 4.4, we examine its compatibility with various architectures and datasets, with further results in Appendix C.4.
- **Sensitivity:** *Is it sensitive to hyperparameters?* In Section 4.6, we analyze the sensitivity of CvLoss to the hyperparameter α , showing stable performance across a broad parameter range.
- **Efficiency:** *What is the computational cost of it?* In Appendix C.8, we evaluate the running cost of CvLoss across different scenarios.

4.1 Setup

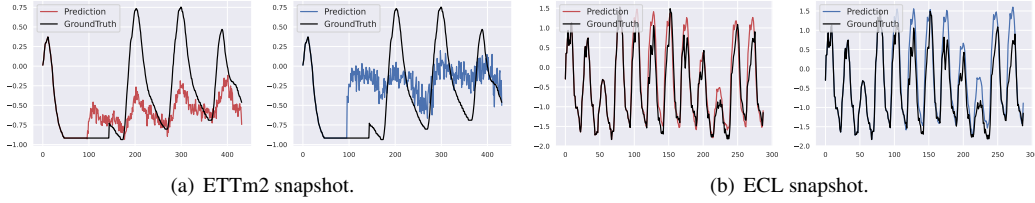
Datasets. Following [Wu et al., 2023], we evaluate on standard benchmarks: four ETT subsets, ECL, Traffic, Weather, and PEMS [Liu et al., 2024]. All datasets are chronologically split into training, validation, and test sets. Statistics are in Appendix B.1.

Baselines. We compare CvLoss to a range of competitive baselines, categorized as: (1) Transformer-based models: PatchTST [Nie et al., 2023], iTransformer [Liu et al., 2024], TQNet [Lin et al., 2025],

Table 1: Overall forecasting performance.

Models	CvLoss (Ours)		TimeFilter (2025)		TQNet (2025)		iTransformer (2024)		Leddiam (2024)		SOFTS (2023)		PatchTST (2023)		Crossformer (2023)		TimesNet (2023)		DLinear (2023)		FEDformer (2022)	
Metrics	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.372	0.381	<u>0.377</u>	<u>0.393</u>	<u>0.377</u>	<u>0.393</u>	0.407	0.410	0.386	0.397	0.393	0.403	0.387	0.400	0.513	0.496	0.400	0.406	0.404	0.408	0.448	0.452
ETTh2	0.419	0.427	<u>0.420</u>	<u>0.428</u>	0.441	0.434	0.454	0.447	0.431	0.429	0.449	0.442	0.469	0.454	0.529	0.522	0.458	0.450	0.461	0.457	0.440	0.460
Weather	0.236	0.260	<u>0.240</u>	0.270	0.242	<u>0.268</u>	0.258	0.279	0.242	0.272	0.255	0.278	0.259	0.281	0.259	0.315	0.259	0.287	0.265	0.315	0.309	0.360
ECL	0.157	0.252	<u>0.159</u>	<u>0.256</u>	0.164	0.258	0.178	0.270	0.169	0.263	0.174	0.264	0.216	0.304	0.244	0.334	0.193	0.295	0.225	0.319	0.214	0.327
Traffic	0.407	0.254	<u>0.408</u>	0.269	0.445	0.276	0.428	0.282	0.467	0.294	<u>0.409</u>	<u>0.267</u>	0.555	0.362	0.550	0.304	0.620	0.336	0.625	0.383	0.610	0.376
Solar	<u>0.218</u>	0.254	0.228	<u>0.262</u>	0.197	0.255	0.233	0.262	0.230	0.264	0.229	0.256	0.270	0.307	0.641	0.639	0.301	0.319	0.330	0.401	0.292	0.381
PEMS03	0.079	0.183	0.084	0.191	<u>0.080</u>	<u>0.186</u>	0.096	0.204	0.101	0.210	0.087	0.192	0.151	0.265	0.138	0.253	0.119	0.271	0.219	0.295	0.167	0.291
PEMS04	0.078	0.181	0.083	0.186	<u>0.081</u>	<u>0.185</u>	0.098	0.207	0.102	0.213	0.091	0.196	0.162	0.273	0.145	0.267	0.109	0.220	0.236	0.350	0.195	0.308
PEMS07	0.063	0.156	0.071	0.170	<u>0.065</u>	<u>0.161</u>	0.088	0.190	0.087	0.192	0.075	0.173	0.166	0.270	0.181	0.272	0.106	0.208	0.241	0.343	0.133	0.282
PEMS08	0.080	0.182	<u>0.083</u>	<u>0.186</u>	0.105	0.203	0.127	0.212	0.102	0.211	0.114	0.208	0.238	0.289	0.232	0.270	0.150	0.244	0.347	0.421	0.234	0.326

Note: We fix the input length as 96 for all baselines following [Liu et al., 2024]. **Bold** and underlined denote best and second-best results, respectively. Avg indicates average results over multiple forecast horizons. CvLoss using TimeFilter [Hu et al., 2025a] as backbone.

Figure 3: The forecast sequence of DF (in blue) and CvLoss (in red), with historical length $H = 96$.

238 Crossformer [Zhang and Yan, 2023], FEDformer [Zhou et al., 2022] and Leddam [Yu et al., 2024];
 239 ((2) Non-transformer models: DLinear [Zeng et al., 2023], SOFTS [Han et al., 2024b], TimesNet [Wu
 240 et al., 2023] and TimeFilter [Hu et al., 2025a].

241 **Implementation.** Baseline implementations closely follow the official codebase from Time-
 242 Filter [Hu et al., 2025a] and TQNet [Lin et al., 2025]. To ensure fair comparison, the drop-last
 243 trick is disabled for all models, as recommended in [Qiu et al., 2024]. All models are trained with
 244 the Adam optimizer [Kingma and Ba, 2015]. When integrating CvLoss into a baseline forecast
 245 model, we retain all hyperparameters from the public benchmarks [Hu et al., 2025a, Lin et al., 2025],
 246 only tuning α and the patch size. Experiments are run on Intel(R) Xeon(R) Platinum 8470Q with 8
 247 NVIDIA RTX 5090 GPUs. Further implementation details are provided in Appendix B.

248 4.2 Overall performance

249 Table 1 compares long- and short-term forecasting results. Adding CvLoss consistently improves
 250 accuracy across datasets; on Traffic, for example, MAE decreases by 0.014. The gains come from
 251 explicit structural regularization over synchronous and asynchronous cross-variable dependencies.

252 **Examples.** A qualitative comparison between forecasts generated by DF versus CvLoss is presented
 253 in Figure 3. While the model trained with DF captures general periodic trends, it often fails to model
 254 accurate amplitudes and intricate temporal dynamics. For example, on the ETTm2 dataset, it suffers
 255 from severe amplitude attenuation, completely missing the prominent ground-truth peaks. Similarly,
 256 on ECL, it overestimates the high periodic peaks, such as those observed around the 1750th and 250th
 257 steps. In contrast, CvLoss accurately captures these large peak magnitudes and complex fluctuations.

Table 2: Comparable results with other learning objectives for time-series forecasting.

Loss		Ours		QDF		Time-o1		FreDF		Koopman		Soft-DTW		DF	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TQNet	ETTm1	0.373	0.386	0.371	0.389	<u>0.372</u>	0.390	0.375	0.390	0.499	0.595	0.394	<u>0.387</u>	0.377	0.393
	ETTh1	0.430	0.438	<u>0.431</u>	0.431	0.432	0.437	0.432	<u>0.432</u>	0.442	0.451	0.453	<u>0.438</u>	0.441	0.434
	ECL	0.162	0.253	<u>0.165</u>	<u>0.257</u>	0.167	<u>0.257</u>	0.168	<u>0.257</u>	0.166	0.258	0.524	0.623	<u>0.165</u>	0.259
	Weather	0.241	0.264	<u>0.242</u>	<u>0.268</u>	0.245	0.269	0.244	<u>0.268</u>	0.306	0.282	0.276	0.255	<u>0.242</u>	<u>0.268</u>
PDF	ETTm1	0.378	0.383	<u>0.381</u>	<u>0.394</u>	0.386	0.399	0.387	0.400	0.587	0.485	0.396	0.404	0.387	0.396
	ETTh1	0.429	<u>0.432</u>	<u>0.436</u>	0.429	0.438	0.438	0.437	0.435	0.497	0.472	0.447	0.447	0.452	0.440
	ECL	0.189	0.269	<u>0.194</u>	0.277	0.195	0.276	<u>0.194</u>	<u>0.274</u>	0.196	0.281	0.695	0.548	0.198	0.281
	Weather	0.259	0.276	0.259	<u>0.281</u>	<u>0.264</u>	0.284	0.268	0.287	0.268	0.290	1.296	0.452	0.265	0.283

Note: **Bold** and underlined denote best and second-best results, respectively. Follow the settings of QDF Wang et al. [2026a].

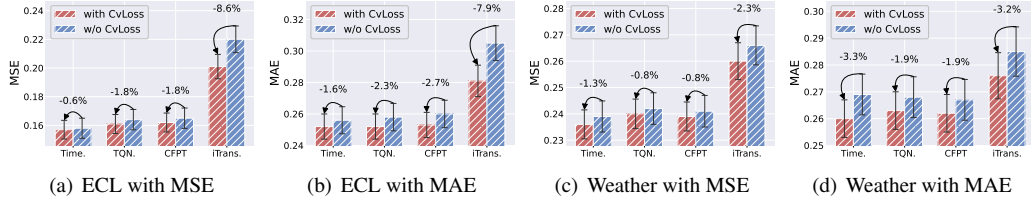


Figure 4: Improvement of CvLoss applied to different forecast models, shown with colored bars for means over forecast lengths (96, 192, 336, 720) and error bars for 50% confidence intervals.

4.3 Learning objective comparison

CvLoss is compared with 11 learning objectives from four categories: shape alignment, likelihood maximization, distribution balancing, and decomposition. Table 2 summarizes average performance on four datasets, with full horizon-level results in Appendix Table 13.

Shape-alignment losses (e.g., Soft-DTW [Cuturi and Blondel, 2017], Koopman [Lange et al., 2021]) and likelihood objectives (e.g., Time-o1 [Wang et al., 2025b], FreDF [Wang et al., 2025a], QDF [Wang et al., 2026a]) improve over MSE, but still treat multivariate outputs as scalar predictions. They therefore leave the spatiotemporal objective gap ($\Delta \neq 0$) in Theorem 3.1 unaddressed. By penalizing edge-wise residual discrepancies, CvLoss ranks first in 20 settings (Appendix Table 13) and generalizes across TQNet [Lin et al., 2025] and PDF [Dai et al., 2024] (Appendix Table 12).

4.4 Generalization studies

We test CvLoss on TimeFilter, TQNet, CFPT [Kou et al., 2025], and iTransformer. As shown in Figure 4, CvLoss improves different architectures and datasets. On ECL, all four backbones improve, with iTransformer reducing MSE by 8.6% and MAE by 7.9%. On Weather, MSE drops by 1.3%–2.3% and MAE by 1.9%–3.3%, showing that CvLoss is broadly plug-and-play.

4.5 Ablation studies

To investigate the impact of cross-variable interaction designs within our proposed CVL, we conduct ablation studies on the construction of the edge set $\mathcal{E}$. Specifically, we compare the following variants:

- CvLoss[†] utilizes a synchronous edge set $\mathcal{E}$ that strictly captures cross-variable interactions at the same time step.
- CvLoss[‡] utilizes an asynchronous edge set $\mathcal{E}$ that exclusively models lagged cross-variable interactions across different time steps (disallowing same-time interactions).
- CvLoss is our default setting, where $\mathcal{E}$ is a fully connected cross-variable graph that permits both synchronous and lagged interactions.

Table 3: Ablation study results.

Model	Sync.	Async.	Data	T=96		T=192		T=336		T=720		Avg	
				MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
DF	$\times$	$\times$	ETTm1	0.311	0.353	0.357	0.378	0.390	0.401	0.449	0.439	0.377	0.393
			ETTh1	0.371	0.393	0.428	0.426	0.475	0.446	<u>0.488</u>	<u>0.470</u>	<u>0.441</u>	<u>0.434</u>
			ECL	0.134	0.229	0.154	0.247	0.170	0.265	0.200	0.293	0.165	0.259
			Weather	0.156	0.200	0.205	0.245	<u>0.262</u>	0.287	0.343	0.342	0.242	0.269
CvLoss [†]	$\checkmark$	$\times$	ETTm1	0.306	0.344	0.357	0.372	<u>0.388</u>	0.399	0.450	0.439	0.375	0.389
			ETTh1	0.369	0.387	<u>0.426</u>	0.422	0.475	0.444	0.505	0.485	0.444	0.435
			ECL	0.134	0.228	<u>0.152</u>	<u>0.243</u>	<u>0.169</u>	<u>0.261</u>	<u>0.199</u>	<u>0.289</u>	0.164	<u>0.255</u>
			Weather	0.159	0.202	0.208	0.246	0.265	0.287	0.344	0.341	0.244	0.269
CvLoss [‡]	$\times$	$\checkmark$	ETTm1	<u>0.305</u>	<u>0.343</u>	<u>0.353</u>	0.373	0.390	0.400	<u>0.448</u>	<u>0.431</u>	<u>0.374</u>	<u>0.387</u>
			ETTh1	<u>0.369</u>	<u>0.387</u>	0.428	<u>0.422</u>	<u>0.473</u>	<u>0.444</u>	0.505	0.485	0.444	0.435
			ECL	<u>0.134</u>	<u>0.228</u>	0.153	0.244	0.170	0.262	0.200	0.290	<u>0.164</u>	0.256
			Weather	<u>0.156</u>	<u>0.196</u>	<u>0.203</u>	<u>0.241</u>	0.263	<u>0.286</u>	<u>0.343</u>	<u>0.340</u>	<u>0.241</u>	<u>0.266</u>
CvLoss	$\checkmark$	$\checkmark$	ETTm1	0.305	0.343	0.352	<u>0.373</u>	0.388	0.400	0.444	0.430	0.372	0.387
			ETTh1	0.368	0.385	0.426	0.422	0.471	0.443	0.486	0.469	0.438	0.430
			ECL	0.133	0.225	0.151	0.241	0.166	0.258	0.196	0.287	0.162	0.253
			Weather	0.154	0.192	0.203	0.238	0.262	0.285	0.342	0.339	0.240	0.264

Note: **Bold** and underlined denote best and second-best results, respectively. "Sync." and "Async." are abbreviations for synchronous and asynchronous.

Table 4: Varying α results of TQNet

α	ETTh2		ECL		Weather	
	MSE	MAE	MSE	MAE	MSE	MAE
0	0.379	0.403	0.164	0.258	0.242	0.269
0.001	0.379	0.403	0.164	0.259	0.242	0.269
0.002	0.379	0.403	0.164	0.258	0.242	0.269
0.005	0.379	0.403	0.164	0.258	0.242	0.268
0.01	0.379	0.403	0.164	0.258	0.242	0.268
0.02	0.379	0.403	0.164	0.258	0.242	0.268
0.05	0.378	0.403	0.164	0.257	0.242	0.268
0.1	0.378	0.402	0.164	0.257	0.242	0.268
0.2	0.377	0.402	<u>0.162</u>	<u>0.255</u>	0.242	0.267
0.5	<u>0.372</u>	<u>0.398</u>	0.162	0.252	<u>0.241</u>	<u>0.265</u>
1	0.365	<u>0.391</u>	0.176	0.264	0.241	0.263

Note: **Bold** and underlined denote the best and second-best results.

Table 5: Varying α results of TimeFilter.

α	ETTh2		ECL		Weather	
	MSE	MAE	MSE	MAE	MSE	MAE
0	0.367	0.399	0.159	0.256	0.241	0.270
0.001	0.367	0.399	0.159	0.256	0.240	0.270
0.002	0.367	0.399	0.159	0.256	0.241	0.270
0.005	0.367	0.399	0.160	0.257	0.241	0.270
0.01	0.367	0.399	0.159	0.256	0.240	0.270
0.02	0.367	0.399	0.163	0.259	0.240	0.270
0.05	0.366	0.399	0.159	0.255	0.240	0.269
0.1	0.366	0.398	0.159	0.255	0.240	0.269
0.2	0.364	0.397	<u>0.159</u>	<u>0.254</u>	<u>0.239</u>	0.267
0.5	0.361	<u>0.393</u>	0.157	0.251	<u>0.237</u>	<u>0.264</u>
1	<u>0.362</u>	0.390	0.168	0.261	0.236	0.260

Note: **Bold** and underlined denote the best and second-best results.

As shown in Table 3, the default CvLoss generally outperforms both CvLoss[†] and CvLoss[‡] in most settings. This confirms that integrating both synchronous and asynchronous interactions within $\mathcal{E}$ is crucial for capturing complex dependencies and yields the best overall forecasting performance.

4.6 Hyperparameter sensitivity

Tables 4 and 5 report the sensitivity of CvLoss to the loss weight α on TQNet and TimeFilter. Increasing α from 0 yields consistent improvements, indicating the benefit of incorporating cross-variable structural dependencies into the learning objective across different backbones and datasets. For example, CvLoss reduces the MSE of TQNet from 0.379 to 0.365 on ETTh2, and that of TimeFilter from 0.159 to 0.157 on ECL. However, using the structural penalty ($\alpha = 1$) is not always optimal; on ECL, both models perform best around $\alpha = 0.5$ and degrade when α reaches 1. This suggests that CvLoss is most effective when combined with point-wise MSE, which anchors forecasting accuracy and prevents translation-invariant solutions under structural regularization.

5 Conclusion

This paper studies the overlooked output-side dependencies in multivariate time series forecasting. We show that the standard point-wise DF objective is mismatched with real-world spatiotemporal correlations, and introduce CvLoss to explicitly regularize cross-variable residual consistency. CvLoss is implemented as a graph total variation penalty over forecast patches, allowing both synchronous and lagged interactions to be modeled without changing the inference procedure. Extensive experiments show that CvLoss consistently improves SOTA forecasting backbones, outperforms competitive learning objectives, and stays efficient in practice. These results suggest that modeling the structural geometry of future variables is a simple but effective direction for improving multivariate forecasting.

References

- Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. itransformer: Inverted transformers are effective for time series forecasting. In *International Conference on Learning Representations*, 2024.
- Lifan Zhao and Yanyan Shen. Rethinking channel dependence for multivariate time series forecasting: Learning from leading indicators. In *The Twelfth International Conference on Learning Representations*, 2024.
- Xiangfei Qiu, Xingjian Wu, Yan Lin, Chenjuan Guo, Jilin Hu, and Bin Yang. Duet: Dual clustering enhanced multivariate time series forecasting. In *SIGKDD*, pages 1185–1196, 2025a.
- Yifan Hu, Guibin Zhang, Peiyuan Liu, Disen Lan, Naiqi Li, Dawei Cheng, Tao Dai, Shu-Tao Xia, and Shirui Pan. Timefilter: Patch-specific spatial-temporal graph filtration for time series forecasting. In *Forty-second International Conference on Machine Learning*, 2025a.
- Hao Wang, Licheng Pan, Zhichao Chen, Degui Yang, Sen Zhang, Yifei Yang, Xinggao Liu, Haoxuan Li, and Dacheng Tao. Fredf: Learning to forecast in the frequency domain. In *ICLR*, 2025a.
- Yifan Hu, Yuante Li, Peiyuan Liu, Yuxia Zhu, Naiqi Li, Tao Dai, Shu tao Xia, Dawei Cheng, and Changjun Jiang. Fintsb: A comprehensive and practical benchmark for financial time series forecasting. *arXiv preprint arXiv:2502.18834*, 2025b.
- Yifan Hu, Peiyuan Liu, Yuante Li, Dawei Cheng, Naiqi Li, Tao Dai, Jigang Bao, and Xia Shu-Tao. Finmamba: Market-aware graph enhanced multi-level mamba for stock movement prediction. *arXiv preprint arXiv:2502.06707*, 2025c.
- Yuxuan Wang, Haixu Wu, Jiaxiang Dong, Yong Liu, Mingsheng Long, and Jianmin Wang. Deep time series models: A comprehensive survey and benchmark. 2024.
- Xiangfei Qiu, Hanyin Cheng, Xingjian Wu, Junkai Lu, Jilin Hu, Chenjuan Guo, Christian S. Jensen, and Bin Yang. A comprehensive survey of deep learning for multivariate time series forecasting: A channel strategy perspective, 2026.
- Xiang Liu, Naiqi Li, and Shu-Tao Xia. Gdtw: A novel differentiable dtw loss for time series tasks. In *Proc. IEEE Int. Conf. Acoust. Speech Signal Process.*, pages 2860–2864. IEEE, 2021.
- Vincent Le Guen and Nicolas Thome. Shape and time distortion loss for training deep time series forecasting models. *Proc. Adv. Neural Inf. Process. Syst.*, 32, 2019.
- Marco Cuturi and Mathieu Blondel. Soft-dtw: a differentiable loss function for time-series. In *Proc. Int. Conf. Mach. Learn.*, pages 894–903. PMLR, 2017.
- Hao Wang, Licheng Pan, Yuan Lu, Zhichao Chen, Tianqiao Liu, Shuting He, Zhixuan Chu, Qingsong Wen, Haoxuan Li, and Zhouchen Lin. Quadratic direct forecast for training multi-step time-series forecast models. In *ICLR*, pages 1–9, 2026a.
- Hao Wang, Licheng Pan, Zhichao Chen, Xu Chen, Qingyang Dai, Lei Wang, Haoxuan Li, and Zhouchen Lin. Time-o1: Time-series forecasting needs transformed label alignment. *Proc. Adv. Neural Inf. Process. Syst.*, 2025b.
- Henning Lange, Steven L Brunton, and J Nathan Kutz. From fourier to koopman: Spectral methods for long-term time series prediction. *Journal of Machine Learning Research*, 22(41):1–38, 2021.
- Licheng Pan, Hao Wang, Haocheng Yang, Yuqi Li, Qingsong Wen, Xiaoxi Li, Zhichao Chen, Haoxuan Li, Zhixuan Chu, and Yuan Lu. Deep time-series forecasting needs kernelized moment balancing, 2026.
- Hao Wang, Licheng Pan, Yuan Lu, Zhixuan Chu, Xiaoxi Li, Shuting He, Zhichao Chen, Haoxuan Li, Qingsong Wen, and Zhouchen Lin. Distdf: Time-series forecasting needs joint-distribution wasserstein alignment. In *Proc. Int. Conf. Learn. Represent.*, pages 1–9, 2026b.
- Xiangfei Qiu, Xingjian Wu, Hanyin Cheng, Xvyan Liu, Chenjuan Guo, Jilin Hu, and Bin Yang. Dbloss: Decomposition-based loss function for time series forecasting. In *NeurIPS*, 2025b.
- Qi Xiong, Kai Tang, Minbo Ma, Ji Zhang, Jie Xu, and Tianrui Li. Modeling temporal dependencies within the target for long-term time series forecasting. *IEEE Transactions on Knowledge and Data Engineering*, 37(12): 7300–7314, 2025. doi: 10.1109/TKDE.2025.3609415.
- Dilfira Kudrat, Zongxia Xie, Yanru Sun, Tianyu Jia, and Qinghua Hu. Patch-wise structural loss for time series forecasting. In *Forty-second International Conference on Machine Learning*, 2025.

353 Yuxuan Liang, Haomin Wen, Yuqi Nie, Yushan Jiang, Ming Jin, Dongjin Song, Shirui Pan, and Qingsong Wen.
354 Foundation models for time series analysis: A tutorial and survey. In *Proceedings of the 30th ACM SIGKDD*
355 *Conference on Knowledge Discovery and Data Mining*, 2024.

356 Si-An Chen, Chun-Liang Li, Sercan O Arik, Nathanael Christian Yoder, and Tomas Pfister. TSMixer: An
357 all-MLP architecture for time series forecast-ing. *Transactions on Machine Learning Research*, 2023. ISSN
358 2835-8856.

359 Yunhao Zhang and Junchi Yan. Crossformer: Transformer utilizing cross-dimension dependency for multivariate
360 time series forecasting. In *International Conference on Learning Representations*, 2023.

361 Haixu Wu, Tengge Hu, Yong Liu, Hang Zhou, Jianmin Wang, and Mingsheng Long. Timesnet: Temporal 2d-
362 variation modeling for general time series analysis. In *International Conference on Learning Representations*,
363 2023.

364 Zonghan Wu, Shirui Pan, Guodong Long, Jing Jiang, Xiaojun Chang, and Chengqi Zhang. Connecting the dots:
365 Multivariate time series forecasting with graph neural networks. In *Proceedings of the 26th ACM SIGKDD*
366 *International Conference on Knowledge Discovery & Data Mining*, 2020.

367 Wenying Han, Tao Zhu, Liming Chen, Huansheng Ning, Yang Luo, and Yaping Wan. Mcformer: Multivariate
368 time series forecasting with mixed-channels transformer. *IEEE Internet of Things Journal*, 11(17):28320–
369 28329, 2024a.

370 TingYu Zhao, Bo Peng, Yuan Sun, DaiPeng Yang, ZhenGuang Zhang, and Xi Wu. Rethinking superpixel
371 segmentation from biologically inspired mechanisms. *Applied Soft Computing*, 156:111467, 2024.

372 Jiawen Wei, Lan Jiang, Pengbo Wei, Ziwen Ye, Teng Song, Chen Chen, and Guangrui Ma. TARFVAE: Efficient
373 one-step generative time series forecasting via TARFLOW based VAE. In *The Thirty-ninth Annual Conference*
374 *on Neural Information Processing Systems*, 2026.

375 Theodoros Evgeniou, Charles A. Micchelli, and Massimiliano Pontil. Learning multiple tasks with kernel
376 methods. *Journal of Machine Learning Research*, 6(21):615–637, 2005.

377 Yuqi Nie, Nam H Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words:
378 Long-term forecasting with transformers. In *International Conference on Learning Representations*, 2023.

379 Shengsheng Lin, Haojun Chen, Haijie Wu, Chunyun Qiu, and Weiwei Lin. Temporal query network for efficient
380 multivariate time series forecasting. In *Forty-second International Conference on Machine Learning*, 2025.

381 Tian Zhou, Ziqing Ma, Qingsong Wen, Xue Wang, Liang Sun, and Rong Jin. FEDformer: Frequency enhanced
382 decomposed transformer for long-term series forecasting. In *International Conference on Machine Learning*,
383 pages 27268–27286. PMLR, 2022.

384 Guoqi Yu, Jing Zou, Xiaowei Hu, Angelica I Aviles-Rivero, Jing Qin, and Shujun Wang. Revitalizing multivariate
385 time series forecasting: Learnable decomposition with inter-series dependencies and intra-series variations
386 modeling. In *Forty-first International Conference on Machine Learning*, 2024.

387 Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In
388 *Proc. AAAI Conf. Artif. Intell.*, 2023.

389 Lu Han, Xu-Yang Chen, Han-Jia Ye, and De-Chuan Zhan. SOFTS: Efficient multivariate time series forecasting
390 with series-core fusion. In *Advances in Neural Information Processing Systems*, 2024b.

391 Xiangfei Qiu, Jilin Hu, Lekui Zhou, Xingjian Wu, Junyang Du, Buang Zhang, Chenjuan Guo, Aoying Zhou,
392 Christian S. Jensen, Zhenli Sheng, and Bin Yang. Tfb: Towards comprehensive and fair benchmarking of
393 time series forecasting methods. In *VLDB*, pages 2363–2377, 2024.

394 Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference*
395 *on Learning Representations*, pages 1–9, 2015.

396 Tao Dai, Beiliang Wu, Peiyuan Liu, Naiqi Li, Jigang Bao, Yong Jiang, and Shu-Tao Xia. Periodicity decoupling
397 framework for long-term series forecasting. *International Conference on Learning Representations*, 2024.

398 Feifei Kou, Jiahao Wang, Lei Shi, Yuhao Yao, Yawen Li, Suguo Zhu, Zhongbao Zhang, and Junping Du.
399 Cfpt: Empowering time series forecasting through cross-frequency interaction and periodic-aware timestamp
400 modeling. In *Forty-second International Conference on Machine Learning*, 2025.

401 Jianxin Li, Xiong Hui, and Wancai Zhang. Informer: Beyond efficient transformer for long sequence time-series
402 forecasting. In *The Thirty-Fifth AAAI Conference on Artificial Intelligence*, 2021.

- 403 Haixu Wu, Jiehui Xu, Jianmin Wang, and Mingsheng Long. Autoformer: Decomposition transformers with
404 Auto-Correlation for long-term series forecasting. In *Advances in Neural Information Processing Systems*,
405 2021.
- 406 Minhao Liu, Ailing Zeng, Muxi Chen, Zhijian Xu, Qiuxia Lai, Lingna Ma, and Qiang Xu. Scinet: time series
407 modeling and forecasting with sample convolution and interaction. In *Advances in Neural Information*
408 *Processing Systems*, 2022.
- 409 Peiyuan Liu, Beiliang Wu, Yifan Hu, Naiqi Li, Tao Dai, Jigang Bao, and Shu-Tao Xia. Timebridge: Non-
410 stationarity matters for long-term time series forecasting. *International Conference on Machine Learning*,
411 2025.

A Theoretical Justification of Theorem 1

In this section, we provide the detailed derivation of Theorem 1, which reveals the suboptimality of the standard Direct Forecasting (DF) objective under real-world spatiotemporal correlations.

Proof. From a probabilistic perspective, finding the optimal parameters θ for the forecasting model f_θ is equivalent to maximizing the likelihood of the ground-truth target Y given the input history X . Let

$$\mathbf{e} = \text{vec}(\hat{Y} - Y) \in \mathbb{R}^{TD}$$

be the flattened forecast error vector.

Assume that the ground-truth residuals follow a zero-mean multivariate Gaussian distribution, i.e.,

$$\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \Sigma_{ST}),$$

where $\Sigma_{ST} \in \mathbb{R}^{TD \times TD}$ is the true spatiotemporal covariance matrix. The probability density function of the residuals is given by:

$$p(\mathbf{e}) = \frac{1}{\sqrt{(2\pi)^{TD} \det(\Sigma_{ST})}} \exp\left(-\frac{1}{2} \mathbf{e}^\top \Sigma_{ST}^{-1} \mathbf{e}\right)$$

Minimizing the Negative Log-Likelihood (NLL) of this distribution yields the optimal objective function. Taking the negative logarithm, we have:

$$-\log p(\mathbf{e}) = \frac{TD}{2} \log(2\pi) + \frac{1}{2} \log \det(\Sigma_{ST}) + \frac{1}{2} \mathbf{e}^\top \Sigma_{ST}^{-1} \mathbf{e}$$

Assuming the covariance matrix Σ_{ST} represents the intrinsic data distribution and is independent of the model parameters θ , the first two terms are constants with respect to the optimization process. By dropping these constants and scaling the objective by $\frac{1}{TD}$ to obtain the average loss per prediction entry, we define the practical NLL objective as:

$$\mathcal{L}_{\text{nll}} = \frac{1}{2TD} \mathbf{e}^\top \Sigma_{ST}^{-1} \mathbf{e}$$

Now, consider the standard point-wise Mean Squared Error (MSE) objective used in the Direct Forecasting (DF) paradigm, defined as:

$$\mathcal{L}_{\text{df}} = \frac{1}{T} \sum_{t=1}^T \frac{1}{D} \|Y_{t,:} - \hat{Y}_{t,:}\|_2^2 = \frac{1}{TD} \mathbf{e}^\top \mathbf{e}$$

Notice that $\mathcal{L}_{\text{df}}$ can be probabilistically interpreted as the negative log-likelihood under a strictly naive assumption: the residuals of all variables across all time steps are entirely independent and share a uniform variance σ^2 . Under this assumption, the covariance matrix reduces to a spherical structure $\Sigma_{ST} = \sigma^2 \mathbf{I}$, where $\mathbf{I} \in \mathbb{R}^{TD \times TD}$ is the identity matrix. In this special case, the NLL naturally degrades to:

$$\mathcal{L}_{\text{nll_naive}} = \frac{1}{2TD} \mathbf{e}^\top (\sigma^2 \mathbf{I})^{-1} \mathbf{e} = \frac{1}{2\sigma^2} \left(\frac{1}{TD} \mathbf{e}^\top \mathbf{e} \right) = \frac{1}{2\sigma^2} \mathcal{L}_{\text{df}}$$

To formally quantify the structural mismatch between the standard DF paradigm and the optimal estimation under true spatiotemporal correlations, we define the structural gap Δ as the difference between the true optimal objective $\mathcal{L}_{\text{nll}}$ and the scaled DF objective $\frac{1}{2\sigma^2} \mathcal{L}_{\text{df}}$:

$$\Delta = \mathcal{L}_{\text{nll}} - \frac{1}{2\sigma^2} \mathcal{L}_{\text{df}} = \frac{1}{2TD} \mathbf{e}^\top \Sigma_{ST}^{-1} \mathbf{e} - \frac{1}{2TD\sigma^2} \mathbf{e}^\top \mathbf{I} \mathbf{e}$$

Factoring out the common terms, we obtain:

$$\Delta = \frac{1}{2TD} \mathbf{e}^\top \left(\Sigma_{ST}^{-1} - \frac{1}{\sigma^2} \mathbf{I} \right) \mathbf{e}$$

In real-world multivariate time series, dynamic physical processes (e.g., disturbance propagation and lagged correlations) dictate that Σ_{ST} is neither diagonal nor uniform. Consequently, $\Sigma_{ST}^{-1} \neq \frac{1}{\sigma^2} \mathbf{I}$, meaning the structural gap $\Delta \neq 0$. Therefore, relying exclusively on $\mathcal{L}_{\text{df}}$ implicitly ignores the off-diagonal elements of the precision matrix Σ_{ST}^{-1} , leading to statistically suboptimal parameter estimation. This concludes the proof.

B Reproduction Details

B.1 Dataset descriptions

Our empirical evaluation is conducted on a diverse collection of widely-used time series forecasting benchmarks. Each dataset presents distinct characteristics in terms of dimensionality and temporal resolution. A summary is provided in Table 6.

Table 6: Dataset description.

Dataset	D	Forecast length	Train / validation / test	Frequency	Domain
ETTh1	7	96, 192, 336, 720	8545/2881/2881	Hourly	Health
ETTh2	7	96, 192, 336, 720	8545/2881/2881	Hourly	Health
ETTm1	7	96, 192, 336, 720	34465/11521/11521	15min	Health
ETTm2	7	96, 192, 336, 720	34465/11521/11521	15min	Health
Weather	21	96, 192, 336, 720	36792/5271/10540	10min	Weather
ECL	321	96, 192, 336, 720	18317/2633/5261	Hourly	Electricity
Traffic	862	96, 192, 336, 720	12185/1757/3509	Hourly	Transportation
Solar-Energy	137	96, 192, 336, 720	36601/5161/10417	10min	Energy
PEMS03	358	12, 24, 36, 48	15617/5135/5135	5min	Transportation
PEMS04	307	12, 24, 48, 96	10172/3375/281	5min	Transportation
PEMS07	883	12, 24, 48, 96	16911/5622/468	5min	Transportation
PEMS08	170	12, 24, 36, 48	10690/3548/265	5min	Transportation

Note: D denotes the number of variates. *Frequency* denotes the sampling interval of time points. *Train*, *Validation*, *Test* denotes the number of samples employed in each split. The taxonomy aligns with [Wu et al., 2023].

- **ETT** [Li et al., 2021]: This dataset includes seven electricity transformer indicators logged between July 2016 and July 2018. It is partitioned into four subsets according to sampling rates: ETTh1 and ETTh2 for hourly data, and ETTm1 and ETTm2 for 15-minute intervals.
- **Weather** [Wu et al., 2021]: Encompasses 21 meteorological factors collected every 10 minutes throughout the year 2020 from the Max Planck Institute for Biogeochemistry.
- **ECL** [Wu et al., 2021]: Records the hourly power usage patterns of 321 individual customers.
- **Traffic** [Wu et al., 2021]: Monitors hourly road occupancy via 862 sensors located on freeways in the San Francisco Bay Area from 2015 to 2016.
- **Solar-Energy**¹: 10-minute solar power production records from 137 photovoltaic plants throughout 2006.
- **PEMS** [Liu et al., 2022]: Derived from public California highway traffic records, aggregated at 5-minute steps. We adopt two standard subsets, PEMS03 and PEMS08, for our study.

Following established protocols [Qiu et al., 2024, Liu et al., 2024], all datasets are segmented chronologically into training, validation, and testing portions. Regarding the ETT, Weather, ECL, Solar and Traffic datasets, we maintain a constant input window of 96 time steps and assess model performance across four distinct forecasting horizons: 96, 192, 336, and 720. For the PEMS datasets, while the historical look-back period remains 96, the evaluation focuses on shorter prediction intervals of 12, 24, 36, and 48 steps. Throughout our experimental phase, we disable the *dropping-last trick* during the final test set assessment to ensure that every data point in the terminal batch is fully utilized.

B.2 Implementation details of model training

To ensure a strictly fair comparison, we build all experiments on top of publicly released benchmark implementations and retain the original model architectures and hyperparameter configurations. When integrating CvLoss, we do not modify the forecasting backbone, data preprocessing, optimizer, training schedule, or evaluation protocol; only the proposed objective term and its associated weight are added. We also disable the drop-last trick for all methods following [Qiu et al., 2024], so that every test sample is evaluated consistently. To establish a fair comparison, we reproduced all baseline models using their official, publicly available implementations, primarily sourcing from the TQNet [Lin et al., 2025], TimeFilter [Hu et al., 2025a], iTransformer [Liu et al., 2024], TimeBridge [Liu et al., 2025] and Time-Series-Library [Wang et al., 2024] repositories. The reproducibility of these baseline results was verified prior to our experiments. All models were trained to minimize the MSE loss function using the Adam optimizer [Kingma and Ba, 2015]. All baselines apply to the publicly available parameters in their library without any changes. To prevent overfitting, we employed an early stopping mechanism that terminates training if the validation loss fails to improve for three consecutive epochs.

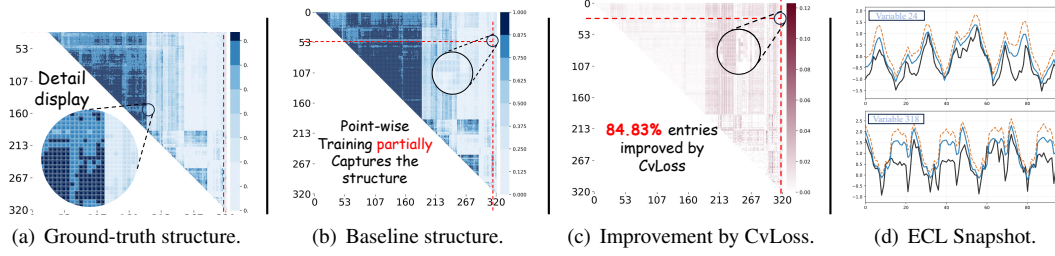


Figure 5: The cross-variable correlation structures of future series given X on the ECL dataset, with input length $H = 96$ and forecast horizon $T = 96$. The correlation matrices are computed from the ground-truth future series (a) and the future series predicted by plain iTransformer [Liu et al., 2024] (b), while (c) shows where iTransformer trained with CvLoss yields smaller structural errors than plain iTransformer, covering 84.83% of cross-variable entries. See details in Appendix B.3.

B.3 Additional Visualization of Cross-Variable Correlation Structures

To further examine how CvLoss affects the recovery of cross-variable dependencies, we provide additional visualizations on the ETTh2 and Weather datasets. These cases follow the same qualitative protocol as Figure 5, while using input length $H = 96$ and forecast horizon $T = 96$.

Let $Y \in \mathbb{R}^{T \times D}$ denote the ground-truth future series and let $\hat{Y}^{\text{base}}, \hat{Y}^{\text{cv}} \in \mathbb{R}^{T \times D}$ denote the future series predicted by the baseline iTransformer and by iTransformer trained with CvLoss, respectively. For any future series $S \in \mathbb{R}^{T \times D}$, we define its cross-variable correlation structure as

$$C(S) = [|\text{Corr}(S_{:,i}, S_{:,j})|]_{i,j=1}^D \in \mathbb{R}^{D \times D},$$

where $\text{Corr}(\cdot, \cdot)$ denotes the Pearson correlation coefficient. Accordingly, we visualize

$$C^{\text{gt}} = C(Y), \quad C^{\text{base}} = C(\hat{Y}^{\text{base}}).$$

To quantify structural deviation from the ground-truth pattern, we define the structural error matrices as

$$E^{\text{base}} = |C^{\text{base}} - C^{\text{gt}}|, \quad E^{\text{cv}} = |C(\hat{Y}^{\text{cv}}) - C^{\text{gt}}|,$$

where the absolute value is taken element-wise. The improvement induced by CvLoss is then measured by

$$\Delta = E^{\text{base}} - E^{\text{cv}}.$$

A positive entry $\Delta_{ij} > 0$ means that CvLoss produces a smaller structural error than the baseline on the (i, j) -th cross-variable entry, i.e., the recovered correlation is closer to the ground-truth structure. Therefore, panel (c) visualizes the positive part of Δ , highlighting where CvLoss improves over the baseline in recovering cross-variable dependencies.

These supplementary cases complement the PEMS03 example in the main text and verify that the structural benefit of CvLoss is not restricted to a single dataset. As shown in Figures 6 and 7, CvLoss consistently improves the recovery of cross-variable correlation structures across different data regimes, including both low-dimensional and higher-dimensional multivariate forecasting benchmarks. This observation is consistent with our main claim that CvLoss acts as a structural regularizer on multivariate outputs and improves the preservation of cross-variable dependencies beyond point-wise supervision alone.

C More Experimental Results

C.1 Overall performance

Additional overall results are reported in Tables 7 and 8, where we provide performance under different forecast horizons.

C.2 Showcase

Additional experimental results of showcases are available in and , where two datasets and two forecast models are involved.

¹<https://www.nrel.gov/grid/solar-power-data.html>

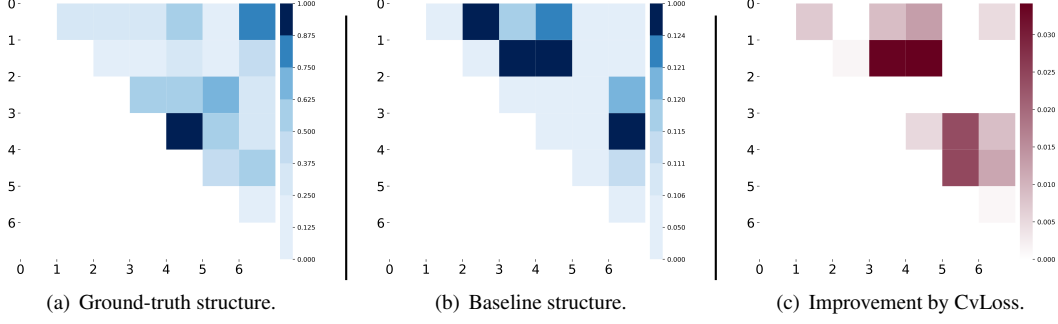


Figure 6: The cross-variable correlation structures of future series given X on the ETTh2 dataset, with input length $H = 96$ and forecast horizon $T = 96$. The correlation matrices are computed from the ground-truth future series (a) and the future series predicted by iTransformer [Liu et al., 2024] (b), while (c) shows the entries on which CvLoss yields ... smaller structural errors than iTransformer (62%).

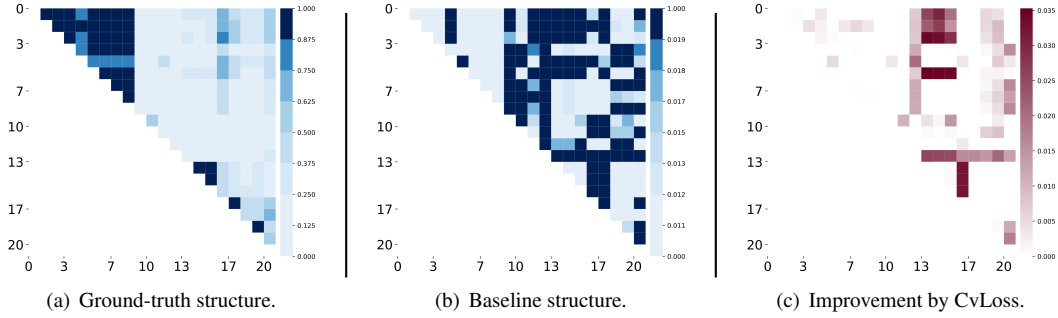


Figure 7: The cross-variable correlation structures of future series given X on the Weather dataset, with input length $H = 96$ and forecast horizon $T = 96$. The correlation matrices are computed from the ground-truth future series (a) and the future series predicted by iTransformer [Liu et al., 2024] (b), while (c) shows the entries on which CvLoss yields smaller structural errors than iTransformer(44%).

504 C.3 Comparison with different learning objectives

505 **Baselines.** Since this paper focuses on the design of learning objectives,, we select competitive learning
 506 objectives tailored for training forecast models as baselines. For the comprehensive evaluation in Table 13, we
 507 compare against 11 objectives, which can be categorized as follows: ❶ **shape-alignment objectives:** GDTW [Liu
 508 et al., 2021], Dilate [Le Guen and Thome, 2019], and Soft-DTW [Cuturi and Blondel, 2017]; ❷ **likelihood**
 509 **maximization objectives:** QDF [Wang et al., 2026a], Time-o1 [Wang et al., 2025b], Koopman [Lange et al.,
 510 2021], FreDF [Wang et al., 2025a], and MSE (also denoted as DF); ❸ **distribution balancing objectives:** KMB-
 511 DF [Pan et al., 2026], and DistDF [Wang et al., 2026b]; and ❹ **decomposition-based objectives:** DBLoss [Qiu
 512 et al., 2025b]. For the cross-model evaluation in Table 12, we select a representative subset of 7 objectives: QDF,
 513 Time-o1, FreDF, Koopman, Soft-DTW, DF, and DBLoss. The implementation of baselines follows the official
 514 codebase from [Pan et al., 2026].

515 **Implementation.** To thoroughly assess the effectiveness of our proposed objective, we evaluate it under two
 516 distinct experimental settings. In the first setting (Table 13), we employ CFPT [Kou et al., 2025] as the default
 517 forecast model to compare all 10 learning objectives across six standard benchmarks (ETTh1, ETTh2, ETTm1,
 518 ETTh2, Weather, and ECL). In the second setting (Table 12), we adopt TQNet and PDF as the backbone models
 519 to verify the generalizability of our method across different architectures. This evaluation is conducted on four
 520 datasets (ETTh1, ETTm1, ECL, and Weather) against the aforementioned subset of 6 learning objectives. To
 521 ensure fair comparison, the drop-last trick is disabled for all baselines, as recommended in [Qiu et al., 2024]. All
 522 objectives are trained with the Adam optimizer [Kingma and Ba, 2015]. When integrating CvLoss to train a
 523 forecast model, we retain all hyperparameters from the public benchmarks [Kou et al., 2025], only tuning α and
 524 the segment patch length.

Table 7: The comprehensive results on the long-term forecasting task.

Models	CvLoss (Ours)		TimeFilter (2025)		TQNet (2025)		iTransformer (2024)		Leddam (2024)		SOFTS (2023)		PatchTST (2023)		Crossformer (2023)		TimesNet (2023)		DLinear (2023)		FEDformer (2022)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.305 0.339	<u>0.313</u> 0.354	<u>0.311</u> <u>0.353</u>	0.334 0.368	0.319 0.359	0.325 0.361	0.329 0.367	0.404 0.426	0.338 0.375	0.346 0.374	0.379 0.419										
	192	0.353 0.367	<u>0.356</u> 0.380	0.357 <u>0.378</u>	0.377 0.391	0.369 0.383	0.375 0.389	0.367 0.385	0.450 0.451	0.374 0.387	0.382 0.391	0.426 0.441										
	336	0.381 0.388	<u>0.386</u> 0.403	0.390 <u>0.401</u>	0.426 0.420	0.394 0.402	0.405 0.412	0.399 0.410	0.532 0.515	0.410 0.411	0.415 0.415	0.445 0.459										
	720	0.450 0.430	<u>0.452</u> <u>0.437</u>	0.449 0.439	0.491 0.459	0.460 0.442	0.466 0.447	0.454 0.439	0.666 0.589	0.478 0.450	0.473 0.451	0.543 0.490										
	Avg.	0.372 0.381	<u>0.377</u> 0.394	<u>0.377</u> <u>0.393</u>	0.407 0.410	0.386 0.397	0.393 0.403	0.387 0.400	0.513 0.496	0.400 0.406	0.404 0.408	0.448 0.452										
ETTh2	96	0.167 0.251	<u>0.169</u> <u>0.255</u>	0.173 <u>0.255</u>	0.180 0.264	0.176 0.257	0.180 0.261	0.175 0.259	0.287 0.366	0.187 0.267	0.193 0.293	0.203 0.287										
	192	0.232 0.294	<u>0.235</u> <u>0.299</u>	0.243 0.300	0.250 0.309	0.243 0.303	0.246 0.306	0.241 0.302	0.414 0.492	0.249 0.309	0.284 0.361	0.269 0.328										
	336	0.290 0.332	<u>0.293</u> <u>0.336</u>	0.301 0.340	0.311 0.348	0.303 0.341	0.319 0.352	0.305 0.343	0.597 0.542	0.321 0.351	0.382 0.429	0.325 0.366										
	720	0.389 0.391	<u>0.390</u> <u>0.393</u>	0.395 0.394	0.412 0.407	0.400 0.398	0.405 0.401	0.402 0.400	1.730 1.042	0.408 0.403	0.558 0.525	0.421 0.415										
	Avg.	0.270 0.317	<u>0.272</u> <u>0.321</u>	0.278 0.322	0.288 0.332	0.281 0.325	0.287 0.330	0.281 0.326	0.757 0.610	0.291 0.333	0.354 0.402	0.305 0.349										
ETTh1	96	0.369 0.391	<u>0.370</u> 0.394	0.371 <u>0.393</u>	0.386 0.405	0.377 0.394	0.381 0.399	0.414 0.419	0.423 0.448	0.384 0.402	0.397 0.412	0.376 0.419										
	192	0.412 0.420	<u>0.413</u> <u>0.420</u>	0.428 0.426	0.441 0.436	0.424 <u>0.422</u>	0.435 0.431	0.460 0.445	0.471 0.474	0.436 0.429	0.446 0.441	0.420 0.448										
	336	0.447 0.439	<u>0.450</u> <u>0.440</u>	0.475 0.446	0.487 0.458	0.459 0.442	0.480 0.452	0.501 0.466	0.570 0.546	0.491 0.469	0.489 0.467	0.459 0.465										
	720	0.447 0.457	<u>0.448</u> <u>0.457</u>	0.488 0.470	0.503 0.491	0.463 <u>0.459</u>	0.499 0.488	0.500 0.488	0.653 0.621	0.521 0.500	0.513 0.510	0.506 0.507										
	Avg.	0.419 0.427	<u>0.420</u> <u>0.428</u>	0.441 0.434	0.454 0.447	0.431 0.429	0.449 0.442	0.469 0.454	0.529 0.522	0.458 0.450	0.461 0.457	0.440 0.460										
ETTh2	96	0.282 0.334	<u>0.283</u> 0.338	0.294 0.344	0.297 0.349	0.292 0.343	0.297 0.347	0.302 0.348	0.745 0.584	0.340 0.374	0.340 0.394	0.358 0.397										
	192	0.356 0.383	<u>0.362</u> 0.392	0.368 0.393	0.380 0.400	0.367 <u>0.389</u>	0.373 0.394	0.388 0.400	0.877 0.656	0.402 0.414	0.482 0.479	0.429 0.439										
	336	0.401 0.419	<u>0.409</u> 0.431	0.417 0.428	0.428 0.432	0.412 <u>0.424</u>	0.410 0.426	0.426 0.433	1.043 0.731	0.452 0.452	0.591 0.541	0.496 0.487										
	720	0.397 0.426	<u>0.408</u> 0.434	0.434 0.446	0.427 0.445	0.419 0.438	0.411 <u>0.433</u>	0.431 0.446	1.104 0.763	0.462 0.468	0.839 0.661	0.463 0.474										
	Avg.	0.359 0.391	<u>0.366</u> 0.399	0.378 0.403	0.383 0.407	0.373 0.399	0.385 0.408	0.387 0.407	0.942 0.684	0.414 0.427	0.563 0.519	0.437 0.449										
Weather	96	0.149 0.188	<u>0.153</u> <u>0.199</u>	0.156 0.200	0.174 0.214	0.156 0.202	0.166 0.208	0.177 0.218	0.158 0.230	0.172 0.220	0.195 0.252	0.217 0.296										
	192	0.198 0.236	<u>0.202</u> 0.246	0.205 <u>0.245</u>	0.221 0.254	0.207 0.250	0.217 0.253	0.225 0.259	0.206 0.277	0.219 0.261	0.237 0.295	0.276 0.336										
	336	0.257 0.283	<u>0.260</u> 0.289	0.262 <u>0.287</u>	0.278 0.296	0.262 0.291	0.282 0.300	0.278 0.297	0.272 0.335	0.280 0.306	0.282 0.331	0.339 0.380										
	720	0.340 0.334	0.345 0.344	<u>0.343</u> <u>0.342</u>	0.358 0.349	0.343 0.343	0.356 0.351	0.354 0.348	0.398 0.418	0.365 0.359	0.345 0.382	0.403 0.428										
	Avg.	0.236 0.260	<u>0.240</u> 0.270	0.242 <u>0.268</u>	0.258 0.279	0.242 0.272	0.255 0.278	0.259 0.281	0.259 0.315	0.259 0.287	0.265 0.315	0.309 0.360										
Electricity	96	0.131 0.225	<u>0.133</u> 0.230	0.134 <u>0.229</u>	0.148 0.240	0.141 0.235	0.143 0.233	0.195 0.285	0.219 0.314	0.168 0.272	0.210 0.302	0.193 0.308										
	192	0.152 0.243	<u>0.154</u> 0.248	<u>0.154</u> <u>0.247</u>	0.162 0.253	0.159 0.252	0.158 0.248	0.199 0.289	0.231 0.322	0.184 0.289	0.210 0.305	0.201 0.315										
	336	0.162 0.257	<u>0.164</u> 0.262	0.170 0.265	0.178 0.269	0.173 0.268	0.178 0.269	0.215 0.305	0.246 0.337	0.198 0.300	0.223 0.319	0.214 0.329										
	720	0.182 0.282	<u>0.184</u> <u>0.284</u>	0.200 0.293	0.225 0.317	0.201 0.295	0.218 0.305	0.256 0.337	0.280 0.363	0.220 0.320	0.258 0.350	0.246 0.355										
	Avg.	0.157 0.252	<u>0.159</u> <u>0.256</u>	0.164 0.258	0.178 0.270	0.169 0.263	0.174 0.264	0.216 0.304	0.244 0.334	0.193 0.295	0.225 0.319	0.214 0.327										
Traffic	96	0.372 0.238	<u>0.375</u> <u>0.251</u>	0.413 0.261	0.395 0.268	0.426 0.276	0.376 <u>0.251</u>	0.544 0.359	0.522 0.290	0.593 0.321	0.650 0.396	0.587 0.366										
	192	0.395 0.248	<u>0.396</u> 0.262	0.432 0.271	0.417 0.276	0.458 0.289	0.398 <u>0.261</u>	0.540 0.354	0.530 0.293	0.617 0.336	0.598 0.370	0.604 0.373										
	336	0.413 0.256	0.416 0.271	0.450 0.277	0.433 0.283	0.486 0.297	<u>0.415</u> <u>0.269</u>	0.551 0.358	0.558 0.305	0.629 0.336	0.605 0.373	0.621 0.383										
	720	0.446 0.275	<u>0.446</u> 0.290	0.486 0.295	0.467 0.302	0.498 0.313	<u>0.447</u> <u>0.287</u>	0.586 0.375	0.589 0.328	0.640 0.350	0.645 0.394	0.626 0.382										
	Avg.	0.407 0.254	<u>0.408</u> 0.269	0.445 0.276	0.428 0.282	0.467 0.294	0.409 <u>0.267</u>	0.555 0.362	0.550 0.304	0.620 0.336	0.625 0.383	0.610 0.376										
Solar-Energy	96	<u>0.183</u> 0.229	0.199 0.239	0.175 0.234	0.203 0.237	0.197 0.241	0.200 <u>0.230</u>	0.234 0.286	0.310 0.331	0.250 0.292	0.290 0.378	0.242 0.342										
	192	<u>0.215</u> 0.250	0.227 0.261	0.198 0.255	0.233 0.261	0.231 0.263	0.229 <u>0.253</u>	0.267 0.310	0.734 0.725	0.296 0.318	0.320 0.398	0.285 0.380										
	336	<u>0.232</u> <u>0.267</u>	0.240 0.272	0.205 0.262	0.248 0.273	0.241 0.268	0.243 0.269	0.290 0.315	0.750 0.735	0.319 0.330	0.353 0.415	0.282 0.376										
	720	<u>0.241</u> 0.270	0.247 0.277	0.210 <u>0.270</u>	0.249 0.275	0.250 0.281	0.245 0.272	0.289 0.317	0.769 0.765	0.338 0.337	0.357 0.413	0.357 0.427										
	Avg.	<u>0.218</u> 0.254	0.228 0.262	0.197 <u>0.255</u>	0.233 0.262	0.230 0.264	0.229 0.256	0.270 0.307	0.641 0.639	0.301 0.319	0.330 0.401	0.292 0.381										
1 st Count		34	39	1	2	<u>6</u>	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Note: We fix the input length as 96 following [Liu et al., 2024]. **Bold** typeface highlights the top performance for each metric, while underlined text denotes the second-best results. Avg indicates the results averaged over forecasting lengths: T=96, 192, 336 and 720.

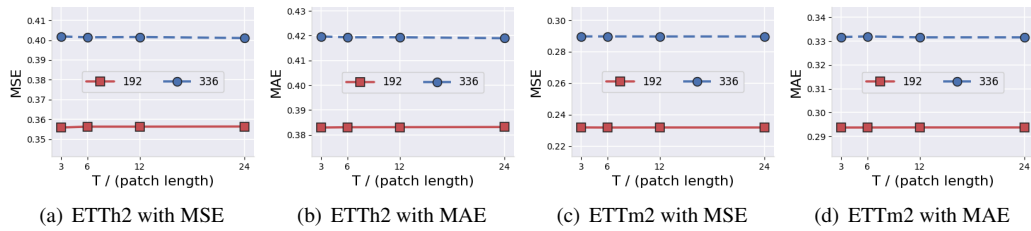


Figure 8: Sensitivity analysis of the patch length hyperparameter. The plots illustrate the forecasting performance in terms of MSE and MAE on the ETTh2 (a, b) and ETTm2 (c, d) datasets. The x-axis represents the ratio of the forecast horizon to the patch length ($T / \text{patch length}$), while the lines denote different forecast horizons ($T = 192$ and $T = 336$). The nearly horizontal curves demonstrate that the predictive accuracy remains consistent regardless of the specific patch length chosen, indicating that CvLoss is highly robust to this hyperparameter.

Table 8: Full results of short-term forecasting.

Models	CvLoss (Ours)	TimeFilter (2025)	TQNet (2025)	iTransformer (2024)	Leddram (2024)	SOFTS (2023)	PatchTST (2023)	Crossformer (2023)	TimesNet (2023)	DLinear (2023)	FEDformer (2022)	
Metric	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	MSE MAE	
PEMS03	12	0.060 0.160	0.063 0.165	<u>0.060 0.161</u>	0.071 0.174	0.068 0.174	0.064 0.165	0.099 0.216	0.090 0.203	0.085 0.192	0.122 0.243	0.126 0.251
	24	0.075 0.180	0.079 0.185	<u>0.077 0.183</u>	0.093 0.201	0.094 0.202	0.083 0.188	0.142 0.259	0.121 0.240	0.118 0.223	0.201 0.317	0.149 0.275
	48	0.101 0.210	0.110 0.222	<u>0.103 0.213</u>	0.125 0.236	0.140 0.254	0.114 0.223	0.211 0.319	0.202 0.317	0.155 0.260	0.333 0.425	0.227 0.348
	Avg.	0.079 0.183	0.084 0.191	<u>0.080 0.186</u>	0.096 0.204	0.101 0.210	0.087 0.192	0.151 0.265	0.138 0.253	0.119 0.271	0.219 0.295	0.167 0.291
PEMS04	12	0.066 0.164	0.068 0.167	<u>0.067 0.166</u>	0.078 0.183	0.076 0.182	0.074 0.176	0.105 0.224	0.098 0.218	0.087 0.195	0.148 0.272	0.138 0.262
	24	0.075 0.178	0.080 0.183	<u>0.077 0.181</u>	0.095 0.205	0.097 0.209	0.088 0.194	0.153 0.257	0.131 0.256	0.103 0.215	0.224 0.340	0.177 0.293
	48	0.094 0.201	0.101 0.209	<u>0.098 0.207</u>	0.120 0.233	0.132 0.249	0.110 0.219	0.229 0.339	0.205 0.326	0.136 0.250	0.335 0.437	0.270 0.368
	Avg.	0.078 0.181	0.083 0.186	<u>0.081 0.185</u>	0.098 0.207	0.102 0.213	0.091 0.196	0.162 0.273	0.145 0.267	0.109 0.220	0.236 0.350	0.195 0.308
PEMS07	12	0.051 0.140	0.055 0.150	<u>0.052 0.144</u>	0.067 0.165	0.066 0.164	0.057 0.152	0.095 0.207	0.094 0.200	0.082 0.181	0.115 0.242	0.109 0.225
	24	0.061 0.155	0.068 0.166	<u>0.063 0.159</u>	0.088 0.190	0.079 0.185	0.073 0.173	0.150 0.262	0.139 0.247	0.101 0.204	0.210 0.329	0.125 0.244
	48	0.077 0.174	0.089 0.193	<u>0.080 0.180</u>	0.110 0.215	0.115 0.228	0.096 0.195	0.253 0.340	0.311 0.369	0.134 0.238	0.398 0.458	0.165 0.288
	Avg.	0.063 0.156	0.071 0.170	<u>0.065 0.161</u>	0.088 0.190	0.087 0.192	0.075 0.173	0.166 0.270	0.181 0.272	0.106 0.208	0.241 0.343	0.133 0.282
PEMS08	12	0.062 0.159	<u>0.064 0.162</u>	0.071 0.170	0.079 0.182	0.070 0.173	0.074 0.171	0.168 0.232	0.165 0.214	0.112 0.212	0.227 0.343	0.173 0.273
	24	0.076 0.178	<u>0.079 0.182</u>	0.095 0.195	0.115 0.219	0.091 0.200	0.104 0.201	0.224 0.281	0.215 0.260	0.141 0.238	0.318 0.409	0.210 0.310
	48	0.101 0.210	<u>0.105 0.214</u>	0.149 0.243	0.186 0.235	0.145 0.261	0.164 0.253	0.321 0.354	0.315 0.335	0.198 0.283	0.497 0.510	0.320 0.394
	Avg.	0.080 0.182	<u>0.083 0.186</u>	0.105 0.203	0.127 0.212	0.102 0.211	0.114 0.208	0.238 0.289	0.232 0.270	0.150 0.244	0.347 0.421	0.234 0.326

Note: The input sequence length is set to 96 for all baselines. All results are averaged across four different forecasting horizons: $T \in \{12, 24, 48\}$. The best and second-best results are highlighted in **bold** and underlined, respectively.

Table 9: The forecasting errors for multivariate time series among CvLoss with four baselines.

Method		TQNet		+ Ours		TimeFilter		+ Ours		CFPT		+ Ours		iTransformer		+ Ours	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh2	96	0.294	0.344	0.281	0.332	0.283	0.337	0.282	0.334	0.287	0.337	0.283	0.333	0.297	0.349	0.296	0.342
	192	0.368	0.393	0.358	0.380	0.362	0.392	0.356	0.383	0.368	0.390	0.361	0.383	0.380	0.400	0.374	0.391
	336	0.417	0.428	0.404	0.419	0.404	0.424	0.401	0.419	0.416	0.428	0.409	0.420	0.428	0.432	0.415	0.427
	720	0.434	0.446	0.413	0.433	0.407	0.433	0.397	0.426	0.415	0.436	0.392	0.418	0.427	0.445	0.421	0.438
	AVG.	0.378	0.403	0.364	0.391	0.364	0.397	0.359	0.391	0.372	0.398	0.361	0.389	0.383	0.407	0.377	0.399
Weather	96	0.156	0.200	0.154	0.192	0.153	0.199	0.149	0.188	0.156	0.201	0.152	0.192	0.174	0.214	0.168	0.207
	192	0.205	0.245	0.203	0.238	0.202	0.246	0.198	0.236	0.205	0.245	0.203	0.240	0.221	0.254	0.221	0.254
	336	0.262	0.287	0.262	0.285	0.260	0.289	0.257	0.283	0.261	0.286	0.260	0.281	0.278	0.296	0.278	0.296
	720	0.343	0.342	0.343	0.339	0.342	0.341	0.340	0.334	0.343	0.339	0.340	0.336	0.358	0.349	0.357	0.347
	AVG.	0.242	0.268	0.241	0.264	0.239	0.269	0.236	0.260	0.241	0.267	0.239	0.262	0.258	0.279	0.256	0.276
ECL	96	0.134	0.229	0.133	0.225	0.133	0.230	0.131	0.225	0.137	0.232	0.136	0.227	0.148	0.240	0.146	0.235
	192	0.154	0.247	0.151	0.241	0.154	0.248	0.152	0.243	0.153	0.247	0.153	0.242	0.162	0.253	0.161	0.249
	336	0.170	0.265	0.166	0.258	0.164	0.261	0.162	0.257	0.168	0.266	0.166	0.259	0.178	0.269	0.174	0.264
	720	0.200	0.293	0.196	0.287	0.184	0.284	0.182	0.282	0.199	0.294	0.196	0.286	0.225	0.317	0.205	0.292
	AVG.	0.164	0.258	0.162	0.253	0.158	0.256	0.157	0.252	0.165	0.260	0.163	0.254	0.178	0.270	0.172	0.260
Wins		0		24		0		24		0		24		0		24	

C.4 Generalization studies

Additional experimental results of varying forecast models are available in Table 9, Table 12 and Fig 9.

C.5 Hyperparameter Sensitivity

We evaluate the stability of CvLoss regarding the loss weight α and patch length L . Loss Weight α : Tables 10 and 11 show accuracy consistently improving as α increases. Optimal performance typically occurs at $\alpha \approx 0.5$ or 1, where structural regularization complements point-wise accuracy. Patch Length L : Figure 8 demonstrates that forecasting performance remains stable across various T/L ratios. The horizontal curves confirm that CvLoss is highly robust to patch size, requiring minimal tuning to achieve significant predictive gains.

C.6 Case study with TQNet and TimeFilter of varying historical lengths

Additional experimental results of varying historical lengths are available in Table 14, complementing the fixed length of 96 used in the main text. The forecast models selected include TimeFilter [Hu et al., 2025a], which is the recent state-of-the-art forecast model, and TQNet [Lin et al., 2025]. The results demonstrate that CvLoss consistently improves both forecast models across different historical sequence lengths.

C.7 Random Seed Sensitivity

Additional experimental results of random seed sensitivity are available in Table 15, where we report the mean and standard deviation of results obtained from experiments conducted with seven different random seeds

Table 10: Varying α results of iTransformer

α	ETTh2		ECL		Weather	
	MSE	MAE	MSE	MAE	MSE	MAE
0	0.384	0.407	0.176	0.267	0.259	0.280
0.001	0.384	0.407	0.176	0.267	0.260	0.281
0.002	0.384	0.407	0.176	0.267	0.260	0.281
0.005	0.384	0.407	0.176	0.267	0.259	0.280
0.01	0.384	0.407	0.176	0.267	0.262	0.282
0.02	0.384	0.407	0.176	0.267	0.261	0.281
0.05	0.384	0.407	0.176	0.267	0.262	0.282
0.1	0.383	0.406	0.175	0.266	<u>0.259</u>	0.280
0.2	0.383	0.406	<u>0.174</u>	<u>0.264</u>	0.259	<u>0.279</u>
0.5	<u>0.379</u>	<u>0.403</u>	0.172	0.260	0.257	0.276
1	0.378	0.399	0.179	0.265	0.258	0.276

Note: **Bold** and underlined denote the best and second-best results.

Table 11: Varying α results of TimeBridge.

α	ETTh2		ECL		Weather	
	MSE	MAE	MSE	MAE	MSE	MAE
0	0.366	0.408	0.159	0.258	0.221	0.261
0.001	0.366	0.408	0.161	0.259	0.221	0.261
0.002	0.366	0.408	0.156	0.255	0.221	0.261
0.005	0.366	0.408	0.159	0.258	0.221	0.261
0.01	0.366	0.408	0.157	0.257	0.221	0.261
0.02	0.366	0.408	0.157	0.257	0.221	0.261
0.05	0.365	0.407	0.157	0.255	0.221	0.261
0.1	0.363	0.406	0.156	0.254	0.220	0.260
0.2	0.360	0.403	0.158	0.255	0.220	0.258
0.5	<u>0.351</u>	<u>0.396</u>	0.153	<u>0.250</u>	0.220	<u>0.255</u>
1	0.342	0.386	<u>0.155</u>	0.248	<u>0.220</u>	0.251

Note: **Bold** and underlined denote the best and second-best results.

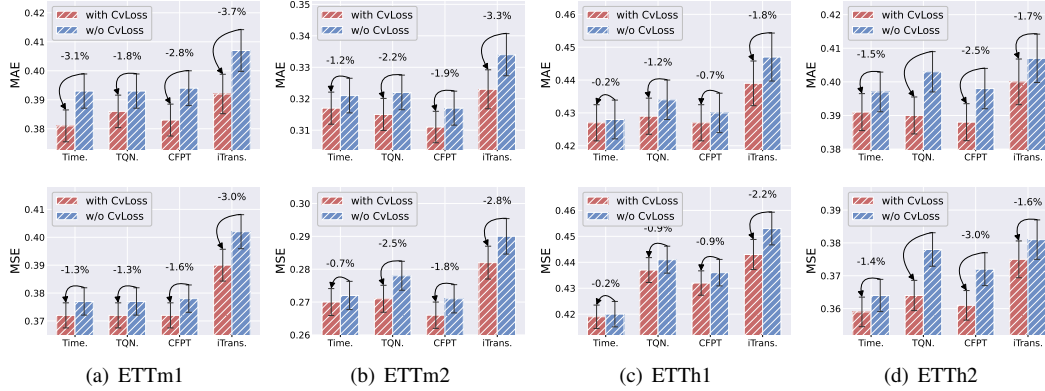


Figure 9: Performance of different forecast models with and without CvLoss. The forecast errors are averaged over forecast lengths and the error bars represent 50% confidence intervals.

(2020–2026). The results indicate minimal sensitivity of the proposed method to random initialization, as all standard deviations remain below 0.004.

C.8 Complexity

We evaluate the computational characteristics of CvLoss from two perspectives: (i) the training and inference overhead introduced by the original formulation, and (ii) the scalability improvement brought by random edge sampling.

We first measure the runtime overhead of CvLoss on ETTh2 and ECL with the batch size fixed to 32. As shown in Fig. 10, CvLoss introduces additional computation only during training, since it regularizes cross-variable dependence through an extra loss term. On the small-scale ETTh2 benchmark, this overhead is marginal in both the forward and backward passes. On the larger ECL benchmark, the overhead becomes more visible, especially in backpropagation, which is expected because modeling cross-variable interactions is more expensive when the number of variables is large. Importantly, although the prediction length T varies from 32 to 1024, the runtime remains in a similar range rather than increasing sharply, indicating that the added cost is well controlled across different horizons.

At inference time, CvLoss is removed from the computation graph and therefore does not participate in prediction. As shown in Fig. 11, the inference latency of the model trained with CvLoss remains highly comparable to that of the baseline iTransformer on both datasets. The ratio plots further show that the latency fluctuates around parity, confirming that CvLoss introduces no practical deployment-time overhead.

While the above results demonstrate that CvLoss is inexpensive at inference and manageable during training, the original formulation still builds a fully connected cross-variable graph. When the number of patches is P and the variable dimension is D , the graph contains $P \times D$ nodes, and the number of pairwise interactions scales as $O(P^2 D^2)$. This quadratic growth becomes a scalability bottleneck on high-dimensional datasets such as ECL. To address this issue, we further introduce *random edge sampling*, which limits the number of evaluated interactions to at most 1000 per batch. In practice, this truncates the cost of CvLoss from rapidly increasing quadratic complexity to an effectively constant level.

Table 12: Comparable results with different learning objectives.

Loss		CvLoss		QDF		Time-o1		FreDF		Koopman		Soft-DTW		DF	
Metrics		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Forecast model:TQNet															
ETTh1	96	0.305	0.343	0.307	0.349	0.309	0.351	0.314	0.355	0.806	0.578	0.315	0.353	0.311	0.353
	192	0.352	0.373	0.352	0.376	0.353	0.375	0.359	0.378	0.619	0.515	0.360	0.377	0.357	0.378
	336	0.388	0.400	0.383	0.398	0.383	0.398	0.382	0.396	0.507	0.468	0.398	0.402	0.390	0.401
	720	0.444	0.430	0.441	0.434	0.444	0.436	0.444	0.432	0.450	0.437	0.476	0.446	0.449	0.439
	Avg	0.373	0.386	0.371	0.389	0.372	0.390	0.375	0.390	0.595	0.499	0.387	0.394	0.377	0.393
ETTh1	96	0.368	0.385	0.365	0.389	0.381	0.395	0.369	0.391	0.415	0.425	0.379	0.390	0.371	0.393
	192	0.426	0.422	0.427	0.421	0.427	0.424	0.425	0.422	0.430	0.422	0.437	0.424	0.428	0.426
	336	0.471	0.443	0.466	0.449	0.471	0.444	0.467	0.445	0.474	0.445	0.488	0.453	0.475	0.446
	720	0.486	0.469	0.466	0.467	0.469	0.466	0.468	0.469	0.483	0.474	0.510	0.487	0.488	0.470
	Avg	0.438	0.430	0.431	0.431	0.437	0.432	0.432	0.432	0.451	0.442	0.453	0.438	0.441	0.434
ECL	96	0.133	0.225	0.135	0.229	0.136	0.228	0.136	0.228	0.137	0.231	0.162	0.258	0.134	0.229
	192	0.151	0.241	0.153	0.245	0.154	0.245	0.155	0.245	0.154	0.247	0.446	0.449	0.154	0.247
	336	0.166	0.258	0.169	0.262	0.171	0.262	0.172	0.263	0.171	0.264	0.912	0.675	0.170	0.265
	720	0.196	0.287	0.202	0.290	0.208	0.293	0.209	0.293	0.204	0.292	0.971	0.715	0.200	0.293
	Avg	0.162	0.253	0.165	0.257	0.167	0.257	0.168	0.257	0.166	0.258	0.623	0.524	0.165	0.259
Weather	96	0.154	0.192	0.158	0.201	0.159	0.201	0.158	0.199	0.223	0.268	0.161	0.202	0.156	0.200
	192	0.203	0.238	0.207	0.245	0.209	0.246	0.209	0.246	0.269	0.304	0.212	0.247	0.205	0.245
	336	0.262	0.285	0.263	0.286	0.268	0.290	0.266	0.288	0.291	0.309	0.270	0.289	0.262	0.287
	720	0.342	0.339	0.342	0.339	0.344	0.341	0.344	0.341	0.346	0.343	0.378	0.365	0.343	0.342
	Avg	0.241	0.264	0.242	0.268	0.245	0.269	0.244	0.268	0.282	0.306	0.255	0.276	0.242	0.268
Forecast model:PDF															
ETTh1	96	0.310	0.342	0.320	0.358	0.326	0.361	0.325	0.362	1.051	0.663	0.323	0.362	0.326	0.363
	192	0.360	0.369	0.361	0.380	0.371	0.386	0.372	0.388	0.420	0.414	0.371	0.388	0.365	0.381
	336	0.390	0.392	0.390	0.401	0.401	0.409	0.399	0.409	0.421	0.415	0.408	0.413	0.397	0.402
	720	0.451	0.430	0.451	0.437	0.448	0.439	0.453	0.443	0.456	0.448	0.480	0.454	0.458	0.437
	Avg	0.378	0.383	0.381	0.394	0.386	0.399	0.387	0.400	0.587	0.485	0.396	0.404	0.387	0.396
ETTh1	96	0.370	0.390	0.375	0.391	0.380	0.403	0.373	0.393	0.632	0.533	0.383	0.405	0.388	0.400
	192	0.419	0.419	0.423	0.419	0.422	0.425	0.423	0.426	0.424	0.429	0.430	0.432	0.440	0.428
	336	0.460	0.439	0.461	0.439	0.463	0.441	0.477	0.446	0.456	0.450	0.462	0.453	0.483	0.449
	720	0.479	0.474	0.484	0.468	0.485	0.483	0.475	0.476	0.476	0.478	0.511	0.496	0.495	0.482
	Avg	0.432	0.429	0.436	0.429	0.438	0.438	0.437	0.435	0.497	0.472	0.447	0.447	0.452	0.440
ECL	96	0.165	0.246	0.171	0.257	0.173	0.253	0.163	0.246	0.194	0.278	0.164	0.250	0.175	0.259
	192	0.174	0.255	0.177	0.261	0.181	0.262	0.179	0.261	0.173	0.260	0.387	0.410	0.182	0.266
	336	0.187	0.269	0.192	0.277	0.196	0.282	0.196	0.278	0.189	0.276	0.966	0.698	0.197	0.282
	720	0.230	0.307	0.234	0.312	0.229	0.307	0.237	0.312	0.228	0.310	1.263	0.834	0.237	0.315
	Avg	0.189	0.269	0.194	0.277	0.195	0.276	0.194	0.274	0.196	0.281	0.695	0.548	0.198	0.281
Weather	96	0.174	0.211	0.176	0.218	0.178	0.219	0.173	0.216	0.202	0.242	0.178	0.219	0.181	0.221
	192	0.224	0.254	0.225	0.260	0.236	0.267	0.235	0.268	0.225	0.258	0.232	0.262	0.234	0.263
	336	0.279	0.294	0.280	0.299	0.284	0.304	0.274	0.295	0.280	0.302	0.281	0.296	0.285	0.300
	720	0.358	0.345	0.357	0.347	0.357	0.348	0.356	0.350	0.353	0.347	4.502	1.036	0.360	0.348
	Avg	0.259	0.276	0.259	0.281	0.264	0.284	0.268	0.287	0.268	0.290	1.296	0.452	0.265	0.283

Fig. 12 compares the training runtime before and after sampling. On ETTh2, where the variable dimension is small, the full CvLoss remains tractable, but the sampled version still makes the runtime flatter as the number of patches increases. On ECL, the benefit is much more pronounced: the runtime of the full CvLoss grows rapidly with the number of patches, whereas the sampled version remains almost unchanged. This confirms that random edge sampling effectively removes the main scalability bottleneck of the original formulation.

Fig. 13 further shows that random edge sampling substantially improves training efficiency without sacrificing predictive performance. As the number of patches increases, the speedup grows rapidly, while the relative MSE change remains close to zero.

Fig. 12 summarizes this observation from an efficiency–accuracy perspective. The results show that large computational gains can be achieved with negligible forecasting degradation, suggesting that a small number of randomly sampled interactions is sufficient to preserve the regularization effect of CvLoss in practice.

Table 13: Full results on the multi-step forecasting task. The length of history window is set to 96 for all baselines. Avg indicates the results averaged over forecasting lengths: $T=96, 192, 336$ and 720 for ETT, ECL, and Weather. Follow the settings of QDF Pan et al. [2026]

Loss	CvLoss (Ours)		KMB-DF (2026)		DBLoss (2025)		QDF (2025)		DistDF (2025)		Time-o1 (2025)		FreDF (2025)		Koopman (2021)		GDTW (2021)		Dilate (2019)		Soft-DTW (2017)		MSE (2002)		
Metrics	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
ETTh1	96	0.309	0.342	0.315	0.352	0.310	0.345	0.318	0.359	0.318	0.355	0.321	0.353	0.320	0.354	0.326	0.363	0.324	0.362	0.323	0.361	0.322	0.359	0.321	0.358
	192	0.350	0.369	0.351	0.379	0.352	0.368	0.355	0.379	0.356	0.380	0.358	0.380	0.356	0.379	0.359	0.383	0.368	0.392	0.363	0.385	0.366	0.385	0.357	0.382
	336	0.385	0.392	0.381	0.400	0.391	0.392	0.384	0.401	0.383	0.401	0.386	0.403	0.386	0.401	0.383	0.401	0.405	0.421	0.384	0.401	0.399	0.409	0.387	0.401
	720	0.442	0.428	0.442	0.433	0.447	0.427	0.443	0.435	0.444	0.434	0.447	0.436	0.444	0.434	0.446	0.436	0.496	0.473	0.446	0.437	0.490	0.461	0.446	0.435
	Avg	0.372	0.383	0.372	0.391	0.375	0.383	0.375	0.393	0.375	0.393	0.378	0.393	0.376	0.392	0.378	0.395	0.398	0.412	0.379	0.396	0.394	0.403	0.378	0.394
ETTh2	96	0.165	0.245	0.166	0.249	0.166	0.244	0.168	0.250	0.168	0.251	0.173	0.251	0.174	0.251	0.182	0.262	0.176	0.257	0.172	0.253	0.176	0.256	0.175	0.256
	192	0.230	0.288	0.229	0.291	0.231	0.288	0.232	0.292	0.232	0.293	0.233	0.291	0.233	0.290	0.235	0.294	0.243	0.305	0.236	0.296	0.243	0.299	0.234	0.295
	336	0.287	0.326	0.289	0.330	0.291	0.327	0.288	0.329	0.289	0.330	0.290	0.329	0.290	0.327	0.291	0.331	0.303	0.344	0.290	0.331	0.306	0.342	0.289	0.330
	720	0.383	0.385	0.384	0.388	0.386	0.387	0.386	0.387	0.385	0.387	0.386	0.385	0.386	0.386	0.388	0.390	0.434	0.419	0.393	0.390	0.438	0.422	0.385	0.389
	Avg	0.266	0.311	0.267	0.315	0.268	0.310	0.268	0.315	0.269	0.315	0.271	0.314	0.271	0.313	0.274	0.319	0.289	0.332	0.273	0.317	0.291	0.330	0.271	0.317
ETTh1	96	0.371	0.387	0.372	0.389	0.370	0.388	0.373	0.392	0.373	0.391	0.369	0.392	0.373	0.389	0.374	0.393	0.377	0.395	0.376	0.394	0.377	0.394	0.373	0.391
	192	0.424	0.418	0.423	0.420	0.424	0.418	0.426	0.421	0.426	0.420	0.424	0.424	0.424	0.418	0.426	0.424	0.434	0.427	0.430	0.423	0.436	0.427	0.427	0.421
	336	0.467	0.441	0.460	0.438	0.466	0.439	0.464	0.441	0.468	0.441	0.464	0.439	0.465	0.439	0.466	0.440	0.486	0.457	0.470	0.442	0.479	0.451	0.466	0.441
	720	0.466	0.461	0.447	0.455	0.468	0.458	0.473	0.464	0.467	0.462	0.462	0.460	0.472	0.468	0.483	0.469	0.512	0.497	0.489	0.478	0.551	0.510	0.477	0.468
	Avg	0.432	0.427	0.426	0.426	0.432	0.426	0.434	0.429	0.434	0.428	0.430	0.429	0.434	0.428	0.437	0.431	0.452	0.444	0.441	0.434	0.461	0.445	0.436	0.430
ETTh2	96	0.283	0.333	0.285	0.337	0.288	0.333	0.288	0.337	0.287	0.337	0.286	0.336	0.289	0.337	0.287	0.338	0.289	0.340	0.287	0.338	0.292	0.342	0.287	0.337
	192	0.361	0.381	0.363	0.387	0.365	0.381	0.367	0.390	0.364	0.391	0.367	0.384	0.365	0.382	0.365	0.390	0.377	0.399	0.368	0.389	0.380	0.398	0.368	0.390
	336	0.409	0.420	0.411	0.425	0.419	0.426	0.412	0.427	0.412	0.426	0.414	0.427	0.414	0.427	0.413	0.429	0.422	0.432	0.413	0.427	0.455	0.443	0.416	0.428
	720	0.392	0.418	0.396	0.426	0.392	0.419	0.401	0.428	0.398	0.427	0.402	0.428	0.406	0.430	0.405	0.430	0.457	0.463	0.415	0.437	0.445	0.453	0.415	0.436
	Avg	0.361	0.388	0.364	0.394	0.366	0.390	0.367	0.396	0.365	0.395	0.367	0.394	0.368	0.394	0.368	0.397	0.386	0.409	0.371	0.398	0.393	0.409	0.372	0.398
ECL	96	0.135	0.227	0.136	0.231	0.135	0.228	0.137	0.232	0.137	0.232	0.141	0.235	0.139	0.232	0.138	0.232	0.882	0.777	0.137	0.232	0.136	0.232	0.137	0.232
	192	0.152	0.242	0.152	0.245	0.152	0.243	0.153	0.246	0.152	0.246	0.154	0.246	0.156	0.246	0.153	0.247	0.853	0.762	0.154	0.247	0.154	0.247	0.153	0.247
	336	0.166	0.259	0.168	0.265	0.165	0.260	0.168	0.264	0.168	0.265	0.168	0.263	0.169	0.263	0.169	0.267	0.809	0.711	0.169	0.267	0.169	0.267	0.168	0.266
	720	0.195	0.285	0.198	0.293	0.193	0.285	0.199	0.293	0.201	0.294	0.198	0.291	0.199	0.290	0.201	0.296	0.900	0.769	0.202	0.297	0.203	0.298	0.199	0.294
	Avg	0.162	0.253	0.163	0.258	0.162	0.254	0.164	0.259	0.164	0.259	0.165	0.259	0.166	0.258	0.165	0.260	0.861	0.755	0.165	0.260	0.165	0.261	0.165	0.260
Weather	96	0.152	0.192	0.152	0.196	0.154	0.193	0.155	0.200	0.155	0.200	0.153	0.197	0.154	0.197	0.173	0.215	0.158	0.206	0.173	0.215	0.177	0.220	0.156	0.201
	192	0.203	0.238	0.204	0.242	0.205	0.239	0.205	0.244	0.205	0.245	0.205	0.243	0.204	0.242	0.205	0.245	0.209	0.250	0.205	0.246	0.209	0.245	0.205	0.245
	336	0.260	0.280	0.260	0.284	0.261	0.280	0.261	0.285	0.261	0.286	0.261	0.284	0.261	0.284	0.262	0.286	0.270	0.296	0.262	0.287	0.278	0.296	0.261	0.286
	720	0.340	0.336	0.339	0.337	0.339	0.333	0.343	0.338	0.343	0.341	0.343	0.340	0.341	0.338	0.345	0.342	0.354	0.352	0.345	0.342	0.385	0.363	0.343	0.339
	Avg	0.239	0.262	0.239	0.265	0.240	0.261	0.241	0.267	0.241	0.268	0.240	0.266	0.240	0.265	0.246	0.272	0.248	0.276	0.246	0.273	0.262	0.281	0.241	0.267
1 st Count	20	20	13	3	7	16	0	0	1	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0

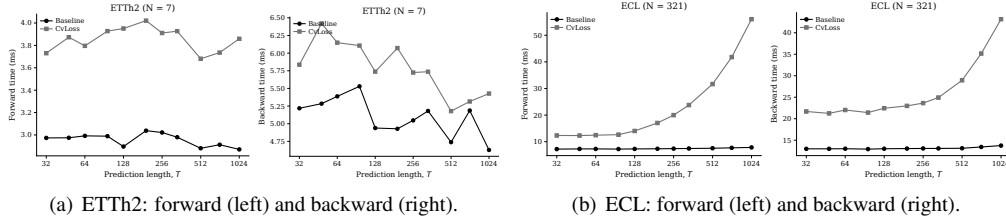


Figure 10: Training latency on ETTh2 and ECL. CvLoss adds little overhead on ETTh2 and a larger, mainly backward-pass overhead on ECL.

577 C.9 Training Weight

578 To deeply understand the adaptive behavior of our proposed learning objective, Table 17 presents a comprehensive
 579 analysis of the learned weights alongside the inherent statistical properties of each dataset, specifically the
 580 average Concordance Correlation Coefficient (CCC) and Pearson Correlation Coefficient.

581 Crucially, we observe a strong positive correspondence between the channel-wise correlation (CCC/Pearson), the
 582 dimensionality (D), and the final assigned weights. For datasets with fewer variables and lower cross-channel
 583 correlation, such as the ETT series ($D = 7$, $CCC \approx 0.14 - 0.18$), the optimization naturally converges to
 584 relatively small and comparable weights for both MSE and the structural penalty, indicating a balanced reliance.

585 Conversely, for high-dimensional spatial-temporal graphs exhibiting strong internal consistency—such as the
 586 PEMS traffic datasets ($D \geq 170$, $CCC > 0.77$) and Solar ($D = 137$, $CCC > 0.91$)—the model adaptively scales
 587 up the magnitude of the learned weights. This phenomenon elegantly demonstrates the data-driven awareness of
 588 our method: when a dataset possesses a rich, highly synchronized cross-variable geometric structure, the model
 589 autonomously assigns larger penalties to explicitly align the complex multivariate dependencies, rather than
 590 relying exclusively on isolated scalar estimations.

Table 14: Varying input sequence length results on the Weather dataset.

Models			CvLoss		TimeFilter		CvLoss		TQNet	
Metrics			MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Historical sequence length	96	96	0.149	0.188	0.153	0.199	0.154	0.191	0.156	0.200
		192	0.198	0.236	0.202	0.246	0.205	0.240	0.205	0.245
		336	0.257	0.283	0.259	0.289	0.263	0.286	0.262	0.287
		720	0.340	0.334	0.345	0.344	0.343	0.339	0.343	0.342
		Avg	0.236	0.260	0.240	0.269	0.241	0.264	0.242	0.268
	192	96	0.142	0.180	0.146	0.195	0.154	0.194	0.153	0.200
		192	0.191	0.230	0.194	0.243	0.198	0.238	0.200	0.244
		336	0.250	0.278	0.256	0.289	0.254	0.283	0.257	0.287
		720	0.333	0.331	0.334	0.341	0.327	0.333	0.326	0.335
		Avg	0.229	0.255	0.232	0.267	0.233	0.262	0.234	0.266
	336	96	0.153	0.192	0.153	0.202	0.148	0.191	0.154	0.205
		192	0.209	0.241	0.215	0.255	0.199	0.242	0.199	0.246
		336	0.269	0.287	0.288	0.302	0.247	0.281	0.249	0.285
		720	0.363	0.347	0.372	0.360	0.319	0.330	0.325	0.337
		Avg	0.248	0.267	0.257	0.280	0.228	0.261	0.232	0.268
	720	96	0.151	0.195	0.159	0.213	0.151	0.198	0.160	0.214
		192	0.216	0.254	0.219	0.266	0.203	0.245	0.210	0.258
		336	0.278	0.301	0.290	0.312	0.262	0.294	0.260	0.299
		720	0.357	0.349	0.392	0.377	0.318	0.332	0.328	0.343
		Avg	0.250	0.275	0.265	0.292	0.233	0.267	0.240	0.278

Table 15: CvLoss ablation on ECL and Weather. Results are reported as mean $\pm$ std over seeds 2020–2026. Lower is better.

Hor.	Met.	ECL				Weather			
		TimeFilter	+CvLoss	TQNet	+CvLoss	TimeFilter	+CvLoss	TQNet	+CvLoss
96	MSE	0.133 $\pm$ 0.001	0.131 $\pm$ 0.000	0.134 $\pm$ 0.000	0.133 $\pm$ 0.000	0.155 $\pm$ 0.002	0.150 $\pm$ 0.001	0.157 $\pm$ 0.001	0.156 $\pm$ 0.002
	MAE	0.230 $\pm$ 0.001	0.225 $\pm$ 0.000	0.230 $\pm$ 0.000	0.225 $\pm$ 0.000	0.202 $\pm$ 0.002	0.189 $\pm$ 0.001	0.200 $\pm$ 0.001	0.194 $\pm$ 0.003
192	MSE	0.154 $\pm$ 0.002	0.152 $\pm$ 0.000	0.153 $\pm$ 0.000	0.151 $\pm$ 0.000	0.204 $\pm$ 0.002	0.201 $\pm$ 0.002	0.205 $\pm$ 0.000	0.204 $\pm$ 0.001
	MAE	0.248 $\pm$ 0.001	0.243 $\pm$ 0.000	0.246 $\pm$ 0.000	0.241 $\pm$ 0.000	0.248 $\pm$ 0.002	0.239 $\pm$ 0.002	0.245 $\pm$ 0.001	0.239 $\pm$ 0.000
336	MSE	0.164 $\pm$ 0.002	0.161 $\pm$ 0.001	0.169 $\pm$ 0.000	0.166 $\pm$ 0.000	0.261 $\pm$ 0.002	0.259 $\pm$ 0.002	0.263 $\pm$ 0.001	0.262 $\pm$ 0.001
	MAE	0.261 $\pm$ 0.002	0.257 $\pm$ 0.000	0.264 $\pm$ 0.000	0.257 $\pm$ 0.000	0.290 $\pm$ 0.001	0.285 $\pm$ 0.002	0.288 $\pm$ 0.001	0.285 $\pm$ 0.001
720	MSE	0.184 $\pm$ 0.003	0.181 $\pm$ 0.001	0.204 $\pm$ 0.004	0.200 $\pm$ 0.003	0.345 $\pm$ 0.001	0.341 $\pm$ 0.002	0.343 $\pm$ 0.001	0.343 $\pm$ 0.000
	MAE	0.284 $\pm$ 0.002	0.282 $\pm$ 0.002	0.297 $\pm$ 0.003	0.292 $\pm$ 0.003	0.344 $\pm$ 0.001	0.336 $\pm$ 0.001	0.342 $\pm$ 0.000	0.339 $\pm$ 0.000
Avg	MSE	0.158 $\pm$ 0.002	0.156 $\pm$ 0.001	0.165 $\pm$ 0.001	0.162 $\pm$ 0.001	0.241 $\pm$ 0.001	0.238 $\pm$ 0.001	0.242 $\pm$ 0.000	0.241 $\pm$ 0.001
	MAE	0.256 $\pm$ 0.001	0.252 $\pm$ 0.001	0.259 $\pm$ 0.001	0.254 $\pm$ 0.001	0.271 $\pm$ 0.001	0.262 $\pm$ 0.001	0.269 $\pm$ 0.000	0.264 $\pm$ 0.001

D Broader Impacts

CvLoss is a general training objective for multivariate time series forecasting. Its potential positive impact is to improve forecasting reliability in applications such as energy management, traffic monitoring, weather analysis, and other operational systems where variables evolve jointly. Better forecasts may support more efficient resource allocation and planning. We do not identify direct negative societal impacts specific to CvLoss, since the method does not introduce new sensitive data, human-subject data, generative capabilities, or deployment-specific decision rules. Any societal risk would mainly come from downstream misuse or misinterpretation of forecasting models in application domains, rather than from the proposed loss itself.

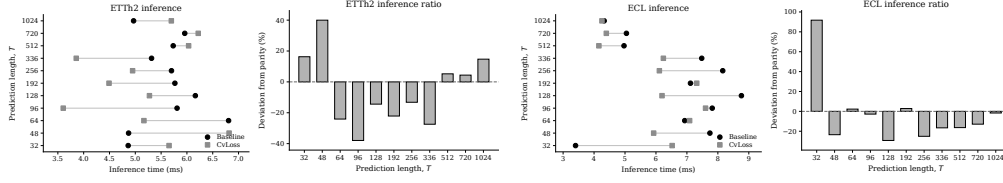
E Limitations

CvLoss uses a predefined cross-variable graph and a fixed patching scheme. Although random edge sampling improves scalability, the graph may still include uninformative interactions, especially for very high-dimensional systems with sparse or domain-specific dependencies. Learning adaptive graph structures and patch sizes could further improve efficiency and interpretability. In addition, this work mainly evaluates deterministic forecasting on regular benchmark datasets; extending CvLoss to probabilistic forecasting, irregular sampling, and missing-value scenarios is left for future work.

Table 16: Comprehensive long-term forecasting results grouped by backbone models in Benchmark TS-Library Wu et al. [2023].

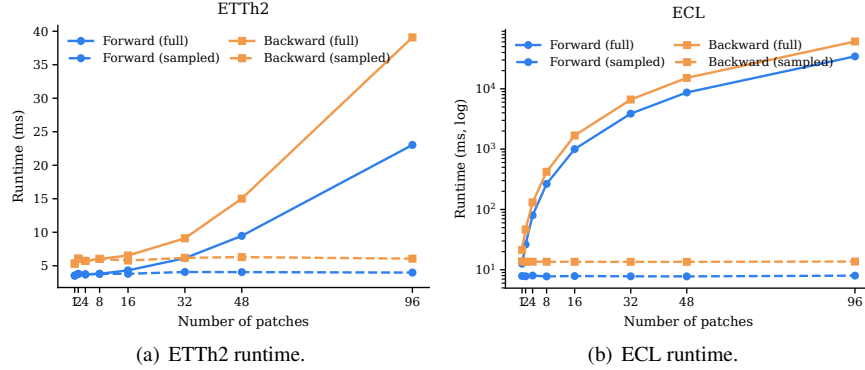
Models	iTransformer						DLinear						PatchTST						
	Original (2024)		DBLoss (2025)		CvLoss (Ours)		Original (2023)		DBLoss (2025)		CvLoss (Ours)		Original (2023)		DBLoss (2025)		CvLoss (Ours)		
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
ETTh1	96	0.347	0.378	<u>0.333</u>	<u>0.356</u>	0.317	0.349	0.346	0.374	<u>0.337</u>	<u>0.359</u>	0.330	0.355	<u>0.324</u>	<u>0.365</u>	0.347	0.368	0.311	0.347
	192	<u>0.375</u>	0.389	0.380	<u>0.381</u>	0.369	0.379	0.382	0.391	<u>0.378</u>	<u>0.380</u>	0.374	0.378	<u>0.372</u>	<u>0.392</u>	0.397	0.394	0.364	0.377
	336	<u>0.409</u>	0.412	0.415	<u>0.404</u>	0.398	0.397	0.415	0.415	<u>0.408</u>	0.401	0.407	<u>0.402</u>	<u>0.398</u>	<u>0.409</u>	0.434	0.415	0.392	0.397
	720	<u>0.477</u>	0.450	0.481	0.441	0.474	<u>0.442</u>	0.473	0.451	0.467	0.437	<u>0.468</u>	<u>0.440</u>	<u>0.457</u>	<u>0.444</u>	0.484	0.446	0.455	0.438
	Avg.	<u>0.402</u>	0.407	<u>0.402</u>	<u>0.395</u>	0.390	0.392	0.404	0.408	<u>0.397</u>	<u>0.394</u>	0.395	0.394	<u>0.388</u>	<u>0.402</u>	0.415	0.406	0.381	0.390
ETTh2	96	0.185	0.271	<u>0.181</u>	<u>0.259</u>	0.176	0.256	0.193	0.293	<u>0.183</u>	0.262	0.180	0.262	0.186	0.268	<u>0.180</u>	<u>0.257</u>	0.178	0.255
	192	0.252	0.312	<u>0.246</u>	<u>0.302</u>	0.241	0.298	0.285	0.361	<u>0.257</u>	<u>0.317</u>	0.246	0.313	<u>0.246</u>	0.305	0.248	<u>0.302</u>	0.240	0.296
	336	0.317	0.353	<u>0.307</u>	<u>0.340</u>	0.302	0.337	0.385	0.429	<u>0.308</u>	<u>0.352</u>	0.305	0.346	0.311	0.348	<u>0.308</u>	<u>0.343</u>	0.300	0.337
	720	0.406	0.402	0.409	0.398	<u>0.407</u>	<u>0.400</u>	0.556	0.523	0.413	0.422	<u>0.414</u>	<u>0.425</u>	0.418	0.414	<u>0.407</u>	<u>0.397</u>	0.402	0.397
	Avg.	0.290	0.334	<u>0.286</u>	<u>0.325</u>	0.282	0.323	0.355	0.402	<u>0.290</u>	<u>0.338</u>	0.286	0.337	0.290	0.334	<u>0.286</u>	<u>0.325</u>	0.280	0.321
ETTh1	96	<u>0.388</u>	0.403	0.389	<u>0.401</u>	0.372	0.394	0.396	0.411	<u>0.391</u>	<u>0.404</u>	0.380	0.389	<u>0.377</u>	<u>0.397</u>	0.383	0.396	0.376	0.394
	192	0.440	0.434	<u>0.439</u>	<u>0.431</u>	0.433	0.427	0.445	0.440	<u>0.440</u>	<u>0.433</u>	0.433	0.425	<u>0.426</u>	0.432	0.429	0.426	0.424	0.427
	336	0.483	0.458	<u>0.479</u>	<u>0.453</u>	0.474	0.451	0.487	0.465	0.478	0.454	<u>0.485</u>	<u>0.458</u>	0.469	<u>0.457</u>	0.462	0.443	<u>0.468</u>	<u>0.453</u>
	720	0.500	0.491	0.473	0.471	<u>0.491</u>	<u>0.484</u>	0.513	0.510	0.489	0.490	<u>0.512</u>	0.509	0.519	0.504	0.452	0.459	<u>0.517</u>	<u>0.499</u>
	Avg.	0.453	0.447	<u>0.445</u>	0.439	0.443	0.439	0.460	0.457	0.449	0.445	<u>0.453</u>	0.445	0.448	0.448	0.431	0.431	<u>0.446</u>	<u>0.443</u>
ETTh2	96	0.297	0.348	<u>0.292</u>	0.341	0.289	0.341	0.341	0.395	<u>0.300</u>	<u>0.351</u>	0.293	0.346	0.309	0.359	0.282	0.333	<u>0.287</u>	<u>0.339</u>
	192	0.379	0.400	<u>0.378</u>	<u>0.394</u>	0.375	0.392	0.482	0.479	<u>0.389</u>	<u>0.408</u>	0.377	0.399	0.381	0.407	0.364	0.386	<u>0.367</u>	<u>0.389</u>
	336	<u>0.422</u>	0.433	0.424	<u>0.430</u>	0.415	0.428	0.593	0.542	<u>0.447</u>	<u>0.453</u>	0.434	0.444	0.412	0.429	0.404	0.419	<u>0.404</u>	0.418
	720	<u>0.428</u>	0.449	0.429	<u>0.444</u>	0.419	0.440	0.840	0.661	0.590	0.539	<u>0.600</u>	0.549	0.436	0.456	0.411	0.433	<u>0.421</u>	<u>0.441</u>
	Avg.	0.381	0.407	<u>0.381</u>	<u>0.402</u>	0.375	0.400	0.564	0.519	<u>0.431</u>	<u>0.438</u>	0.426	0.434	0.384	0.413	0.365	0.393	<u>0.370</u>	<u>0.396</u>
Weather	96	<u>0.184</u>	<u>0.224</u>	0.191	<u>0.224</u>	0.176	0.211	<u>0.196</u>	0.256	0.201	0.245	0.195	<u>0.246</u>	<u>0.176</u>	0.218	0.179	<u>0.213</u>	0.173	0.211
	192	<u>0.231</u>	<u>0.262</u>	0.242	0.268	0.223	0.253	0.239	0.299	<u>0.236</u>	0.278	0.235	<u>0.288</u>	<u>0.221</u>	0.256	0.224	<u>0.253</u>	0.219	0.251
	336	<u>0.286</u>	0.302	0.288	<u>0.298</u>	0.279	0.293	<u>0.281</u>	0.331	0.280	0.312	0.285	<u>0.329</u>	0.280	0.298	<u>0.278</u>	0.291	<u>0.277</u>	<u>0.293</u>
	720	<u>0.363</u>	0.352	0.364	<u>0.349</u>	0.360	0.347	0.345	0.382	0.339	0.359	<u>0.345</u>	<u>0.379</u>	0.356	0.349	<u>0.355</u>	0.342	0.355	<u>0.343</u>
	Avg.	<u>0.266</u>	<u>0.285</u>	0.271	<u>0.285</u>	0.260	0.276	0.265	0.317	0.264	0.299	<u>0.265</u>	<u>0.310</u>	<u>0.258</u>	0.280	0.259	<u>0.275</u>	0.256	0.274
Electricity	96	<u>0.192</u>	<u>0.277</u>	0.200	0.280	0.164	0.248	<u>0.210</u>	0.302	0.211	<u>0.296</u>	0.195	0.275	<u>0.180</u>	<u>0.273</u>	0.185	0.274	0.180	0.270
	192	<u>0.201</u>	<u>0.289</u>	0.206	0.290	0.183	0.264	<u>0.210</u>	0.305	<u>0.210</u>	<u>0.299</u>	0.194	0.278	<u>0.187</u>	<u>0.280</u>	0.192	0.281	0.187	0.276
	336	<u>0.221</u>	<u>0.308</u>	0.225	0.309	0.192	0.276	<u>0.223</u>	0.319	<u>0.223</u>	<u>0.313</u>	0.206	0.292	<u>0.204</u>	<u>0.296</u>	0.209	0.297	0.204	0.293
	720	<u>0.267</u>	<u>0.344</u>	0.271	0.345	0.263	0.337	0.258	0.350	<u>0.257</u>	<u>0.342</u>	0.244	0.329	<u>0.246</u>	<u>0.328</u>	0.252	0.331	0.243	0.325
	Avg.	<u>0.220</u>	<u>0.305</u>	0.226	0.306	0.201	0.281	0.225	0.319	<u>0.222</u>	<u>0.310</u>	0.210	0.293	<u>0.204</u>	<u>0.294</u>	0.209	0.296	0.204	0.291
Traffic	96	<u>0.545</u>	<u>0.371</u>	0.597	0.386	0.434	0.285	0.697	0.429	<u>0.690</u>	<u>0.421</u>	0.642	0.379	<u>0.459</u>	0.298	0.461	<u>0.292</u>	0.455	0.290
	192	<u>0.560</u>	<u>0.380</u>	0.595	0.386	0.469	0.309	0.647	0.407	<u>0.639</u>	<u>0.399</u>	0.595	0.361	<u>0.469</u>	0.301	0.470	<u>0.296</u>	0.465	0.295
	336	<u>0.591</u>	<u>0.395</u>	0.620	0.400	0.465	0.297	0.653	0.410	<u>0.645</u>	<u>0.400</u>	0.601	0.358	<u>0.483</u>	0.307	0.483	<u>0.301</u>	0.478	0.298
	720	<u>0.657</u>	<u>0.427</u>	0.686	0.433	0.568	0.370	0.694	0.429	<u>0.682</u>	<u>0.417</u>	0.647	0.397	<u>0.517</u>	0.326	0.518	<u>0.319</u>	0.512	0.315
	Avg.	<u>0.588</u>	<u>0.393</u>	0.624	0.401	0.484	0.315	0.673	0.419	<u>0.664</u>	<u>0.409</u>	0.621	0.374	<u>0.482</u>	0.308	0.483	<u>0.302</u>	0.478	0.300

Note: Average results (Avg) are calculated over $T \in \{96, 192, 336, 720\}$. Evaluation for the **best** and **second-best** results are independently conducted within the **iTransformer**, the **DLinear**, and the **PatchTST**. "Ours" refers to different enhancements applied to the respective base models.



(a) ETTh2: inference latency (left) and latency ratio (right). (b) ECL: inference latency (left) and latency ratio (right).

Figure 11: Inference latency on ETTh2 and ECL. CvLoss remains comparable to the baseline, with ratios close to parity.



(a) ETTh2 runtime.

(b) ECL runtime.

Figure 12: Training runtime versus the number of patches. Dashed lines denote random edge sampling with at most 1000 edges.

F Statement on the Use of Large Language Models (LLMs)

In accordance with the conference guidelines, we disclose our use of Large Language Models (LLMs) in the preparation of this paper as follows:

We used LLMs (specifically, OpenAI GPT-5.2, GPT-5.4 and Google Gemini 3) *solely for checking grammar errors and improving the readability of the manuscript*. The LLMs were not involved in research ideation, the development of research contributions, experiment design, data analysis, or interpretation of results. All substantive content and scientific claims were created entirely by the authors. The authors have reviewed all LLM-assisted text to ensure accuracy and originality, and take full responsibility for the contents of the paper. The LLMs are not listed as an author.

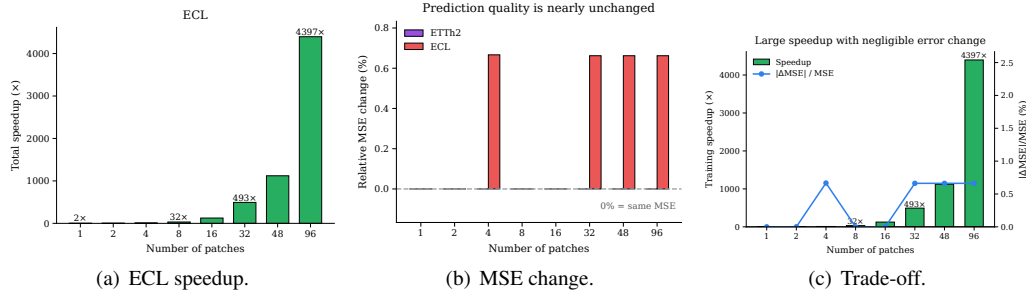


Figure 13: Optimization benefit of random edge sampling. The speedup grows rapidly with the number of patches, while the relative MSE change remains close to zero. The trade-off plot summarizes that large speedups are achieved with negligible performance degradation.

Table 17: The learned loss weights and channel-wise correlations across datasets. The datasets are ordered by their number of variables (D).

Dataset	Channels (D)	Avg CCC	Avg Pearson	Weight for MSE	Weight for Add Loss
ETTh1	7	0.1765	0.2516	1.2706	1.2657
ETTh2	7	0.1449	0.4001	1.2080	1.2002
ETTm1	7	0.1804	0.2601	2.3572	1.9831
ETTm2	7	0.1436	0.3952	2.1759	2.0209
Weather	21	0.2079	0.3520	2.4887	2.2864
Solar	137	0.9115	0.9157	6.9919	3.4617
PEMS08	170	0.7799	0.8161	17.1380	4.2828
PEMS04	307	0.7739	0.7858	18.0127	4.3956
ECL	321	0.4619	0.4965	8.1102	3.0385
PEMS03	358	0.8206	0.8431	21.7118	4.8293
Traffic	862	0.5289	0.5643	4.6804	3.0653
PEMS07	883	0.7810	0.8066	22.8313	4.9570

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#).

Justification: The abstract and Introduction state the paper’s scope, motivation, main objective-gap claim, and CvLoss contribution. The Method, Experiments, and Appendix Limitations sections provide the corresponding theoretical, empirical, and scope details.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#).

Justification: The Appendix includes a separate Limitations section. It discusses the predefined graph, fixed patching scheme, scalability considerations, and evaluation scope.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes].

Justification: The main theoretical result states the residual distribution and covariance assumptions in Theorem 3.1. A complete derivation is provided in the Appendix section on theoretical justification.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes].

Justification: Section 4.2 and Appendix B describe the datasets, splits, baselines, optimizer, early stopping, and evaluation protocol. The abstract also provides an anonymized code URL.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.

- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes].

Justification: The paper provides an anonymized code link in the abstract and uses standard public benchmark datasets cited in the paper. Dataset descriptions and reproduction details are included in Appendix B.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes].

Justification: The experimental setup specifies datasets, data splits, input and forecast lengths, optimizer, early stopping, baseline implementations, and hyperparameter handling. Additional details are provided in Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes].

Justification: The paper reports 50% confidence intervals in the model-generalization figures and $\text{mean}_{\pm\text{std}}$ results over seeds 2020–2026 in Table 15. These results provide variability information for the main empirical claims.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes].

Justification: The paper states the CPU and GPU type used for experiments in Section 4.2, and the appendix includes a Compute Resources section. Appendix C.8 further reports training and inference runtime analyses for CvLoss.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes].

Justification: The work uses public benchmark time-series datasets and standard model training procedures, and does not involve human subjects or high-risk data release. We are not aware of any aspect that violates the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes].

Justification: The appendix includes a Broader Impacts section discussing potential positive impacts in forecasting applications. It also states that we do not identify direct negative societal impacts specific to CvLoss.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A].

Justification: The paper does not release high-risk models, scraped datasets, generative models, or systems with an obvious dual-use misuse risk. The released asset is an implementation of a forecasting loss on public benchmarks.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes].

Justification: The paper cites the datasets and baseline implementations used in the experiments. The appendix further states that the necessary permissions have been obtained and that the experiments comply with the corresponding licenses and terms of use.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes].

Justification: The paper releases an anonymized implementation link and documents the method, training protocol, datasets, and reproduction details in the main text and appendix. The new asset is code rather than a new dataset or model checkpoint.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A].

Justification: The paper does not involve crowdsourcing, user studies, or research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A].

Justification: The paper does not involve crowdsourcing, user studies, or research with human subjects, so IRB approval or equivalent review is not applicable.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

917 • Depending on the country in which research is conducted, IRB approval (or equivalent) may be
 918 required for any human subjects research. If you obtained IRB approval, you should clearly state
 919 this in the paper.

920 • We recognize that the procedures for this may vary significantly between institutions and
 921 locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for
 922 their institution.

923 • For initial submissions, do not include any information that would break anonymity (if applica-
 924 ble), such as the institution conducting the review.

925 **16. Declaration of LLM usage**

926 Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard
 927 component of the core methods in this research? Note that if the LLM is used only for writing, editing,
 928 or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the
 929 research, declaration is not required.

930 Answer: [N/A].

931 Justification: LLMs are not used as an important, original, or non-standard component of the core
 932 method. The appendix separately discloses that LLMs were used only for grammar checking and
 933 readability improvement.

934 Guidelines:

935 • The answer [N/A] means that the core method development in this research does not involve
 936 LLMs as any important, original, or non-standard components.

937 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not be
 938 described.